

SAMA7D6 Series System-in-Package (SiP) MPUs with up to 4-Gbit DDR SDRAM

SAMA7D6 Series SiP Data Sheet



[Product Page Links](#)

Scope

This document provides an overview of the main features of the SAMA7D6 Series SiP microprocessors.

Reference documents for product information on the SAMA7D6 Series microprocessors and DDR2/DDR3L SDRAM memories are listed in the [Reference Documents](#) section.

Introduction

The SAMA7D6 Series SiP integrates the Arm® Cortex®-A7 processor-based SAMA7D6 Series MPU with up to 4-Gbit DDR SDRAM. By combining the SAMA7D6 with DDR SDRAM in a single package, PCB routing complexity, area and number of layers are reduced in most cases. This makes board design easier and more robust by facilitating design for EMI, ESD and signal integrity.

The SAMA7D6 Series SiP is available in the following pin-to-pin compatible SDRAM memory types and densities:

- 512-Mbit DDR2 SDRAM
- 1-Gbit DDR3L SDRAM
- 2-Gbit DDR3L SDRAM
- 4-Gbit DDR3L SDRAM

Reference Documents

Type	Title	Available	Literature No.
Data sheet	SAMA7D6 Series	www.microchip.com	DS60001851
Errata sheet	SAMA7D6 Series Silicon Errata and Data Sheet Clarification	www.microchip.com	DS80001131
Application note	SAMA7D65 Hardware Design Considerations	www.microchip.com	AN5822
Data sheet	512-Mbit 8M × 4 BANKS × 16 BIT DDR2 SDRAM	www.winbond.com	W9751G6NB25I
Data sheet	1-Gbit 8M × 8 BANKS × 16 BIT DDR3L SDRAM	www.winbond.com	W631GU6NB12I
Data sheet	2-Gbit 16M × 8 BANKS × 16 BIT DDR3L SDRAM	www.winbond.com	W632GU6NB12I
Data sheet	4-Gbit 32M × 8 BANKS × 16 BIT DDR3L SDRAM	www.winbond.com	W634GU6NB12I

1. Features

- Arm Cortex-A7 Core
 - Arm TrustZone®
 - Arm® Neon™ multimedia architecture
 - Floating-Point Unit
 - Embedded Trace module with instruction trace stream, including 16 Kbytes of CoreSight™ Embedded Trace buffer
 - 32 Kbytes of L1 data cache, 32 Kbytes of L1 instruction cache
 - 256 Kbytes of L2 cache
 - Up to 1 GHz operational frequency
 - Voltage and frequency scaling support
 - 64-bit generic timers
- Memories
 - Internal memory architecture
 - 128 Kbytes of SRAM
 - 80 Kbytes of ROM, embedding a secure boot loader (boot on QSPI NOR, SLC/MLC NAND, SD card, eMMC)
 - 96 Kbytes ROM for NAND Flash ECC tables
 - 40 Kbytes ROM for crypto libraries (RSA, ECC, etc.)
 - 11 Kbytes of internal OTP
 - 16-bit high-bandwidth DDR2 or DDR3L SDRAM up to 533 MHz and up to 4 Gbits
 - External memory support
 - 16-bit static memory controller, FPGA and memory-mapped peripheral support with synchronous clock
- System
 - Power-on reset cells, reset controller, shutdown controller, Watchdog and secure Watchdog Timers running on internal slow RC oscillator (32 kHz typical) and real-time clock running on slow crystal oscillator (32.768 kHz)
 - Two internal trimmed RC oscillators with typical values: 32 kHz and 12 MHz
 - Two crystal oscillators: 32.768 kHz and 20 to 50 MHz
 - Nine PLLs for core, system bus and peripherals, serial interfaces, DDR I/Os, pixel clock, audio, USB and Ethernet
 - Two 32-channel DMAs with per-channel security configuration
 - One 8-channel DMA dedicated to memory-to-memory transactions
 - Eight programmable clock output signals
- Power Considerations
 - Different power domains and power modes to reduce power consumption
 - Low-power consumption in Backup mode with 5 Kbytes of secure backup SRAM and DDR SDRAM in Self-Refresh mode
 - Low-power with SRAM and register retention, wake-up from various events (USB, CAN, Ethernet WOL, FLEXCOMs), internal events (RTC, timer) and I/O activity
 - Embedded LDOs for analog and PLLs to enable low-cost power management solutions

- Optimum connection to Microchip MCP16501/2 PMICs to enter and exit various power modes of the application
- Multimedia Peripherals
 - Audio
 - Two synchronous serial controllers, each with up to eight channels of up to 32-bit TDM data
 - Two inter-IC sound multi-channel controllers with TDM256 support
 - Up to two 4-channel pulse density microphone controllers; support for eight microphones in parallel
 - One Sony/Philips digital interface transmitter and receiver
 - Audio sample rate converter including four stereo channels
- Display Subsystem Supporting One Display Up to WXGA/720p (1366x768p60/1280x720p60)
 - One 4-lane MIPI D-PHY interface and one DSI controller
 - One single-channel LVDS interface, and one LVDS controller
 - 8-bit databus serial RGB interface
 - One LCD controller with one base layer, one overlay and one high-end overlay
 - One 2D GPU controller with up to WXGA/720p target display resolution and support for RGB and YUV
- Connectivity
 - USB Subsystem:
 - Two high-speed USB devices and three high-speed USB hosts sharing three on-chip transceivers
 - Two USB Type-C™ controllers (TCPC)
 - Two 10/100/1000 Gigabit Ethernet MACs supporting:
 - RGMII and RMII interfaces
 - Energy efficiency as per IEEE 802.3az
 - Ethernet AVB support with IEEE802.1AS timestamping
 - IEEE802.1Qav credit-based traffic shaping hardware support
 - IEEE1588 Precision Time Protocol
 - IEEE1588 Timestamp Unit with TSU timer comparison signal triggering a timer counter and available on a PIO line
 - Support for traffic scheduling and Time-Sensitive Networking (TSN/AVB)
 - Packet buffer support
 - Five flexible data rate CAN-FD controllers with SRAM-based mailboxes with time- and event-triggered transmission, flexible data rate, and a 32-bit timestamp unit
 - Eleven FLEXCOMs supporting U(S)ART, SPI, TWI/I2C with FIFOs
 - I3C controller interface supporting FM, FM+, SDR and HDR-DDR modes
- Mass Storage
 - One 8-bit high-speed memory card host with eMMC 5.1 (HS400), SD3.0, SDR104 mode support
 - One 4-bit high-speed memory card host with eMMC 5.1 (HS200), SD3.0, SDR104 mode support
 - One 4-bit high-speed memory card host with SD3.0, SDR104 mode support

- One octal serial peripheral interface supporting high-speed DDR mode
- One quad serial peripheral interface
- 8-bit SLC and MLC NAND controller with up to 32-bit Error Correcting Code (ECC)
- General-Purpose Analog and Digital Peripherals
 - Six 64-bit periodic interval timers
 - Two three-channel 32-bit timer counters with PWM generation
 - One four-channel 16-bit PWM controller
 - One 19-channel, 12-bit, 1 Msps analog-to-digital converter
 - One analog comparator with four differential inputs
- Safety
 - Zero-power power-on reset cells
 - Main crystal monitor and clock failure detector with failsafe switchover to main RC oscillator
 - 32 kHz crystal monitor and clock failure detector with switch to internal 32 kHz RC
 - Integrity check monitor based on SHA256
 - One Watchdog Timer running on RC oscillator
 - Register write protection
- Security
 - TrustZone support
 - One secure TrustZone Watchdog Timer running on RC oscillator, providing protection against TrustZone starvation
 - Temperature, voltage and frequency monitoring (also applicable for safety purposes)
 - Secure backup SRAM
 - 5 Kbytes scrambled with non-imprinting support powered with VBAT or VDDIN33:
 - 1 Kbyte non-erasable on tamper detection
 - 4 Kbytes erasable on tamper detection
 - Four tamper pins for static or dynamic detection
 - Can be used as regular wake-up lines
 - 256-bit backup register, erasable on tamper detection (tamper pins or monitor outputs); time stamping of tamper events
 - Programmable OTP with bits available for user purposes
 - Configurable JTAG/SWD security (full debug, non-secure-only debug, no debug)
 - Two independent 128-bit AES on-the-fly encryption/decryption on DDR memory, SMC, QSPI0 and QSPI1, including automatic key load at start-up; separate key for secure and non-secure accesses (TZAESB)
 - Secure RTC
- Cryptography
 - Physically Unclonable Function with automatic key load to hardware encryption engines
 - SHA (SHA1, SHA224, SHA256, SHA384, SHA512) compliant with FIPS Publications 180-2
 - AES: 256-, 192-, 128-bit key algorithm, compliant with FIPS PUB 197 specifications
 - TDES: Two-key or three-key algorithms, compliant with FIPS PUB 46-3 specifications
 - True random number generator with health tests compliant with NIST Special Publication 800-22 Tests Suite and FIPS PUB 140-2 and 140-3

- Public Key Coprocessor (CPKCC) and associated Classical Public Key Cryptography Library (CPKCL) for RSA, DSA, ECC GF(2n), ECC GF(p) and associated library for RSA4096, ECC521
- I/O Ports
 - Up to 142 general-purpose I/Os
 - Fully programmable through Set/Clear registers
 - Multiplexing of up to eight peripheral functions per I/O line
 - Each I/O line can be assigned to a peripheral or used as a general-purpose I/O
 - Synchronous output, possibility to set or clear up to 32 I/O lines simultaneously in a single write
 - General-purpose analog and digital inputs are tolerant to positive and negative current injection
- Design for Low Electromagnetic Interference (EMI)
 - Slew rate controlled I/Os
 - DDR Phy with impedance-calibrated drivers
 - Programmable spread spectrum PLLs
 - Careful BGA power/ground ball assignment to provide optimum decoupling capacitors placement
- Microchip Recommended Power Management Integrated Circuits (PMICs)
 - MCP16502, 6-channel PMIC with I²C control interface; supports dynamic voltage scaling and processor Low-Power modes (BSR)
 - MCP16501, 4-channel PMIC optimized for compact PCB layout; supports processor Low-Power mode (BSR)
- Operating Conditions
 - Junction temperature range (T_J): -40°C to +105°C
- Packages
 - 427-ball TFBGA 21x18 mm, 0.8 mm pitch
 - 442-ball TFBGA 16x16 mm, 0.5 mm pitch

2. Ordering Information

For details on ordering codes, refer to the [Product Identification System](#) section.

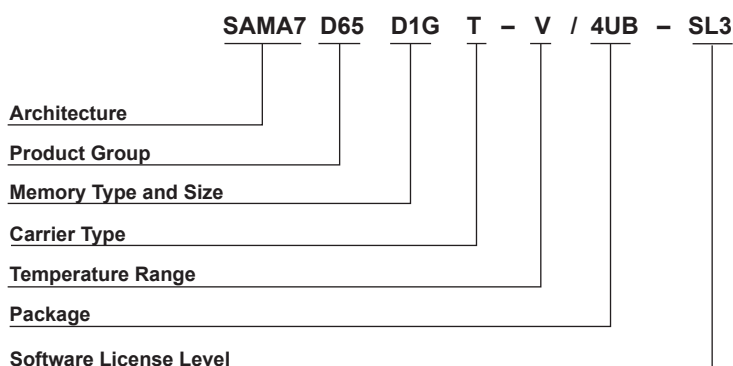
Ordering Code ⁽¹⁾	Memory Type	Tier	Max CPU Speed	Package	Junction Temperature Range ⁽²⁾
SAMA7D65D5M(T)-V/4UB(-SLx)	DDR2 SDRAM	Industrial	1 GHz	TFBGA427	-40°C to +105°C
SAMA7D65D1G(T)-V/4UB(-SLx)	DDR3L SDRAM				
SAMA7D65D2G(T)-V/4UB(-SLx)					
SAMA7D65D4G(T)-V/4UB(-SLx)					
SAMA7D65D1G(T)-V/XHB(-SLx)					
SAMA7D65D2G(T)-V/XHB(-SLx)					
SAMA7D65D4G(T)-V/XHB(-SLx)					
				TFBGA442	

Notes:

1. For SL1, SL2 and SL3 device availability, contact a Microchip Sales representative.
2. Applies to both the MPU and the DDR memory junction temperatures.

3. Product Identification System

To order or obtain information, for example, on pricing or delivery, contact Microchip: <https://www.microchip.com/en-us/about/contact-us>.



Architecture:	SAMA7	= Arm Cortex-A7 CPU
Product Group:	D65	= General-purpose microprocessors with graphic features
Memory Type and Size:	D5M	= 512-Mbit DDR2 SDRAM
	D1G	= 1-Gbit DDR3L SDRAM
	D2G	= 2-Gbit DDR3L SDRAM
	D4G	= 4-Gbit DDR3L SDRAM
Carrier Type:	Blank	= Standard packaging (tray)
	T	= Tape and Reel
Temperature Range (Junction):	V	= -40°C to +105°C (industrial)
Package:	4UB	= TFBGA427
	XHB	= TFBGA442
Software License Level:	Blank	= Standard
	SL1	= Level 1
	SL2	= Level 2
	SL3	= Level 3

Example:

- SAMA7D65D1GT-V/4UB-SL3 = Arm Cortex-A7 general-purpose microprocessor, 1-Gbit DDR3L SDRAM, tape and reel packaging, industrial temperature, 427-ball TFBGA, software license level 3

Note: The Tape and Reel identifier and the Software License Level identifier only appear in the catalog part number description. These identifiers are used for ordering purposes and are not printed on the device package. Check with your Microchip Sales Office for package availability.

4. DDR2 SDRAM Features

The SAMA7D6 Series SiP is available with a 512-Mbit DDR2 SDRAM memory option.

For power consumption, electrical characteristics and memory timings, refer to the manufacturer's documentation listed in the [Reference Documents](#) section.

- Power supply: DDRM_VDD = 1.8V $\pm$ 0.1V
- Double Data Rate architecture: two data transfers per clock cycle
- CAS latency: 3
- Burst length: 8
- Bidirectional, differential data strobes (DQS and DQSN) are transmitted/received with data
- Edge-aligned with read data and center-aligned with write data
- DLL aligns DQ and DQS transitions with clock
- Differential clock inputs (CLK and CLKN)
- Data masks (DM) for write data
- Commands entered on each positive CLK edge; data and data mask referenced to both edges of DQS
- Auto-Refresh and Self-Refresh modes
- Precharged power-down and active power-down
- Write data mask
- Write latency = read latency - 1 (WL = RL - 1)

5. DDR3L SDRAM Features

The SAMA7D6 Series SiP is available with 1-Gbit, 2-Gbit or 4-Gbit DDR3L SDRAM memory options. For power consumption, electrical characteristics and timings of these memories, refer to the manufacturers' data sheets listed in the [Reference Documents](#) section.

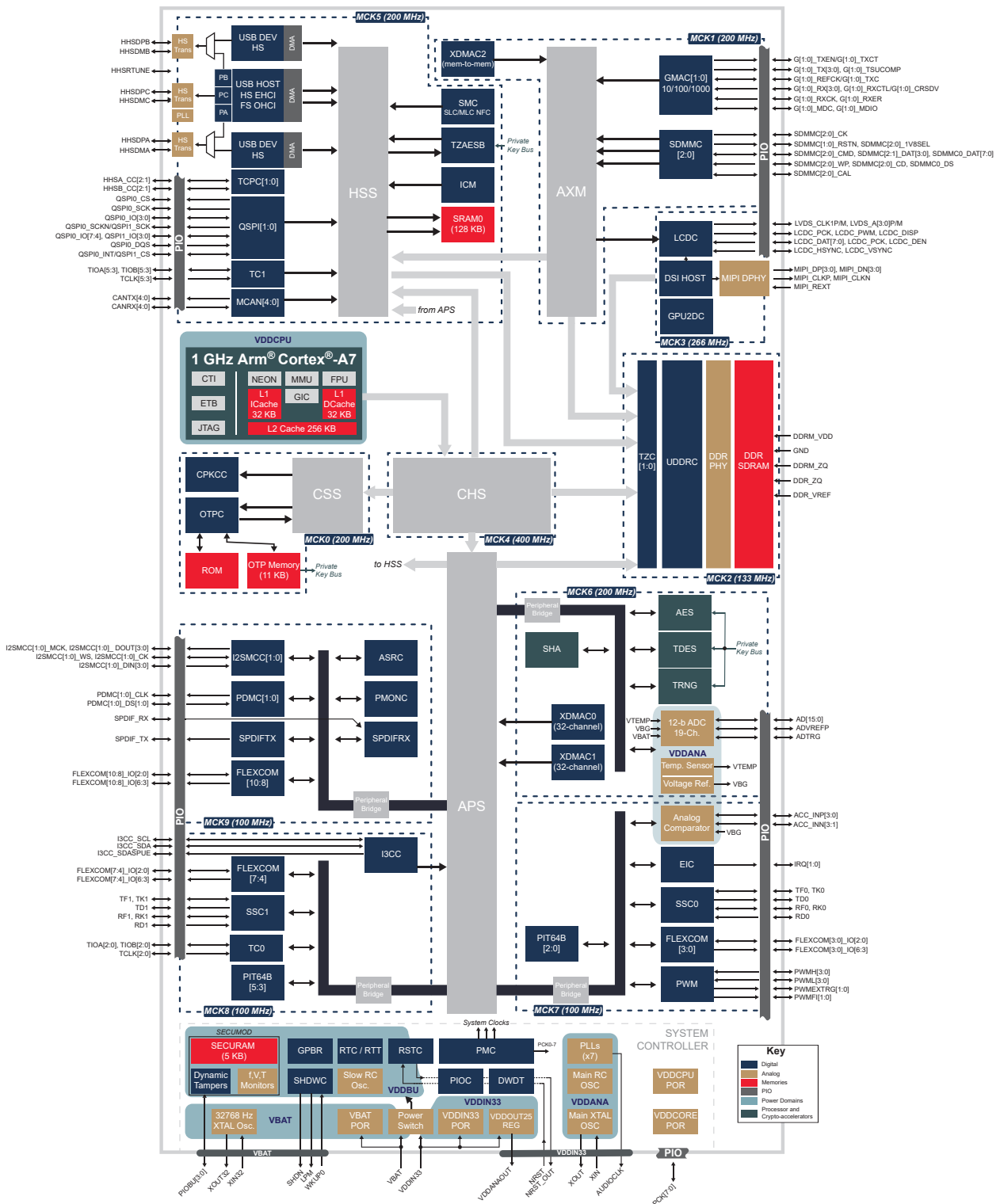
- Power supply: DDR3L DDRM_VDD = 1.283V to 1.45V
- 2-Kbyte page size (x16)
- 8-bank operation controlled by BA0, BA1 and BA2
- Burst lengths (BL): 8 and 4 with Burst Chop (BC)
- Precharge: auto-precharge option for each burst access
- Refresh: auto-refresh, self-refresh
- Average refresh period:
 - 7.8 μ s at $-40^{\circ}\text{C} \leq T_j \leq +85^{\circ}\text{C}$
 - 3.9 μ s at $+85^{\circ}\text{C} < T_j \leq +95^{\circ}\text{C}$
 - 1.95 μ s at $+95^{\circ}\text{C} < T_j \leq +105^{\circ}\text{C}$
- High-speed data transfer realized by the 8-bit prefetch pipelined architecture
- Double Data Rate architecture: two data transfers per clock cycle
- Bidirectional differential data strobe (DQS and /DQS) transmitted/received with data for capturing data at the receiver
- DQS edge-aligned with data for reads and center-aligned with data for writes
- Differential clock inputs (CK and /CK)
- DLL aligns DQ and DQS transitions with CK transitions
- Commands entered on each positive CK edge; data and data mask referenced to both edges of DQS
- Data mask (DM) for write data
- Posted CAS by programmable additive latency for improved command and databus efficiency
- MultiPurpose Register (MPR) for predefined pattern readout
- ZQ calibration for DQ drive and ODT
- Programmable Partial Array Self-Refresh (PASR)
- Reset pin for power-up sequence and reset function
- Self-Refresh Temperature (SRT) range: normal/extended
- Automatic Self-Refresh (ASR)
- Programmable output driver impedance control

6. Configuration Summary

Feature	SAMA7D65D5M	SAMA7D65D1G	SAMA7D65D2G	SAMA7D65D4G
CPU	Cortex-A7 up to 1 GHz			
Package	TFBGA427	TFBGA427	TFBGA427	TFBGA427
		TFBGA442	TFBGA442	TFBGA442
Internal SDRAM (16-bit, 533 MHz)	512-Mbit DDR2 SDRAM	1-Gbit DDR3L SDRAM	2-Gbit DDR3L SDRAM	4-Gbit DDR3L SDRAM
NAND Flash Controller	8-bit			
Ethernet MAC with TSN support	2x 10/100/1000 with RMII/RGMII I/F			
USB Device/Host	3 USB interfaces: 2 host/device with Type-C interface and 1 host			
2D GPU	One controller to accelerate 2D graphic display			
LCD	1 up to WXGA/720p resolution, single-channel LVDS, 4-lane MIPI DSI® Support of 8-bit serial RGB interface			
Number of PIOs	142			
Quad/Octal SPI	2 quad SPI or 1 octal SPI			
SDIO/SD/e.MMC	3			
SMC	16-bit data/15-bit address/2 CS			
CAN-FD	5			
FLEXCOM (USART/UART/I ² C/SPI)	11			
I3C	1 controller			
ADC Inputs	16 external + 3 internal			
ACC Inputs	4 external			
64-bit Periodic Interval Timer	6			
Timer Counter	6 channels			
PWM	4 channels			
ASRC	4x stereo channels with independent conversion rates			
I2SMCC	2 (up to 8 TX and 8 RX channels each)			
SSC	2 (up to 16 TX and 16 RX channels each)			
PDMC	2 (up to 8 digital microphones)			
Cryptography	Asymmetric cryptography (CPKCC+CPKCL), AES, SHA, TRNG, TDES, PUF			

7. Block Diagram

Figure 7-1. SAMA7D6 Series SiP Block Diagram



8. Chip Identifier

Table 8-1. Chip ID and Extended ID Definition

Chip Name	CHIPID_CIDR	CHIPID_EXID ⁽¹⁾
SAMA7D65D5M	0x8026211x	0x00000010 0x000000C8 (SL1) 0x000000C9 (SL2) 0x000000CA (SL3)
SAMA7D65D1G		0x00000018 0x000000CC (SL1) 0x000000CD (SL2) 0x000000CE (SL3)
SAMA7D65D2G		0x00000020 0x000000D0 (SL1) 0x000000D1 (SL2) 0x000000D2 (SL3)
SAMA7D65D4G		0x00000028 0x000000D4 (SL1) 0x000000D5 (SL2) 0x000000D6 (SL3)

Note:

1. For more information, refer to [Product Identification System](#).

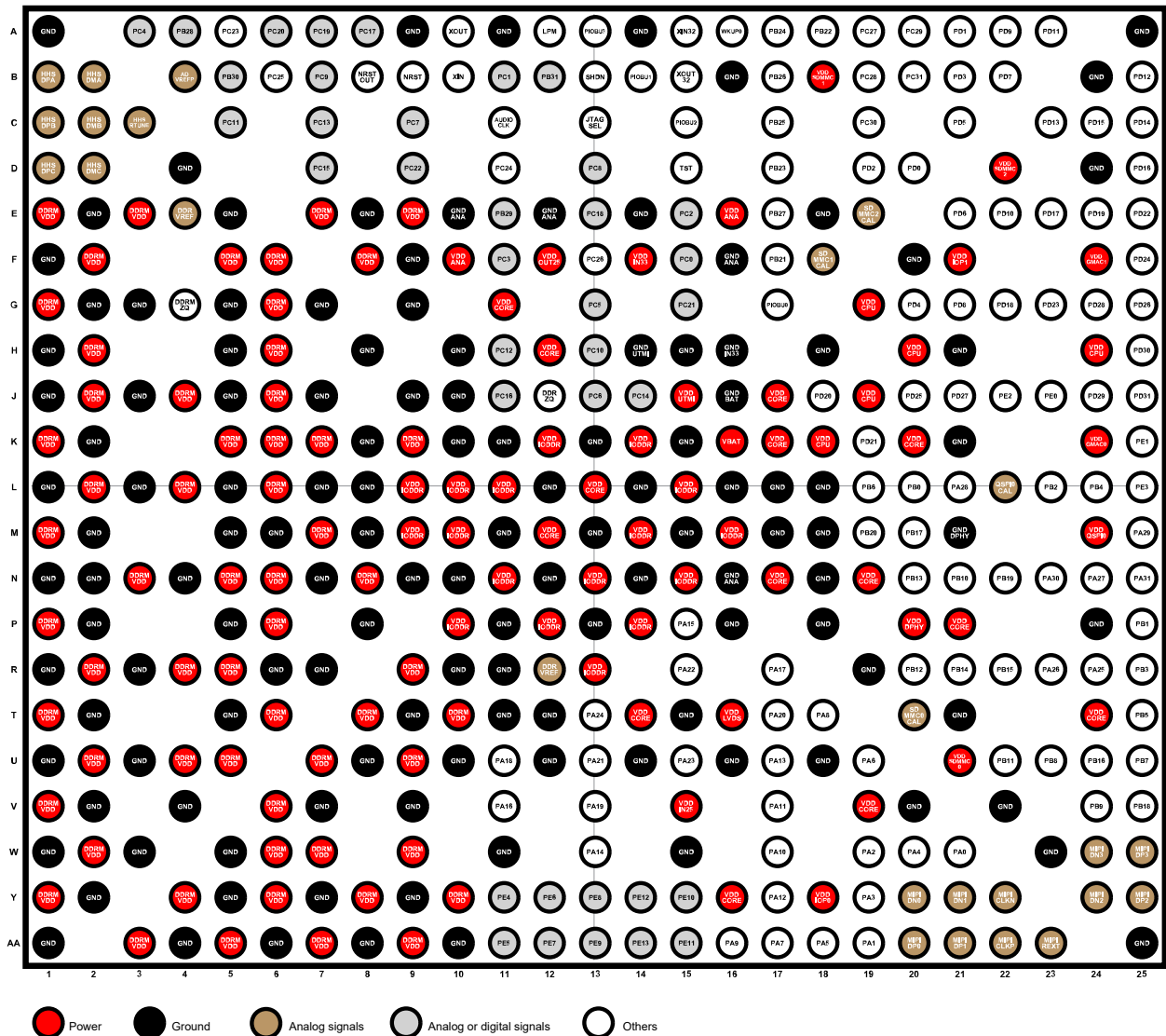
9. Package and Pinout

All SAMA7D6 Series SiP devices using the same package are pin-to-pin compatible.

Package Name	Ball Count	Ball Pitch	Package Size
TFBGA427	427	0.8 mm	21 x 18 x 1.2 mm
TFBGA442	442	0.5 mm	16 x 16 x 1.2 mm

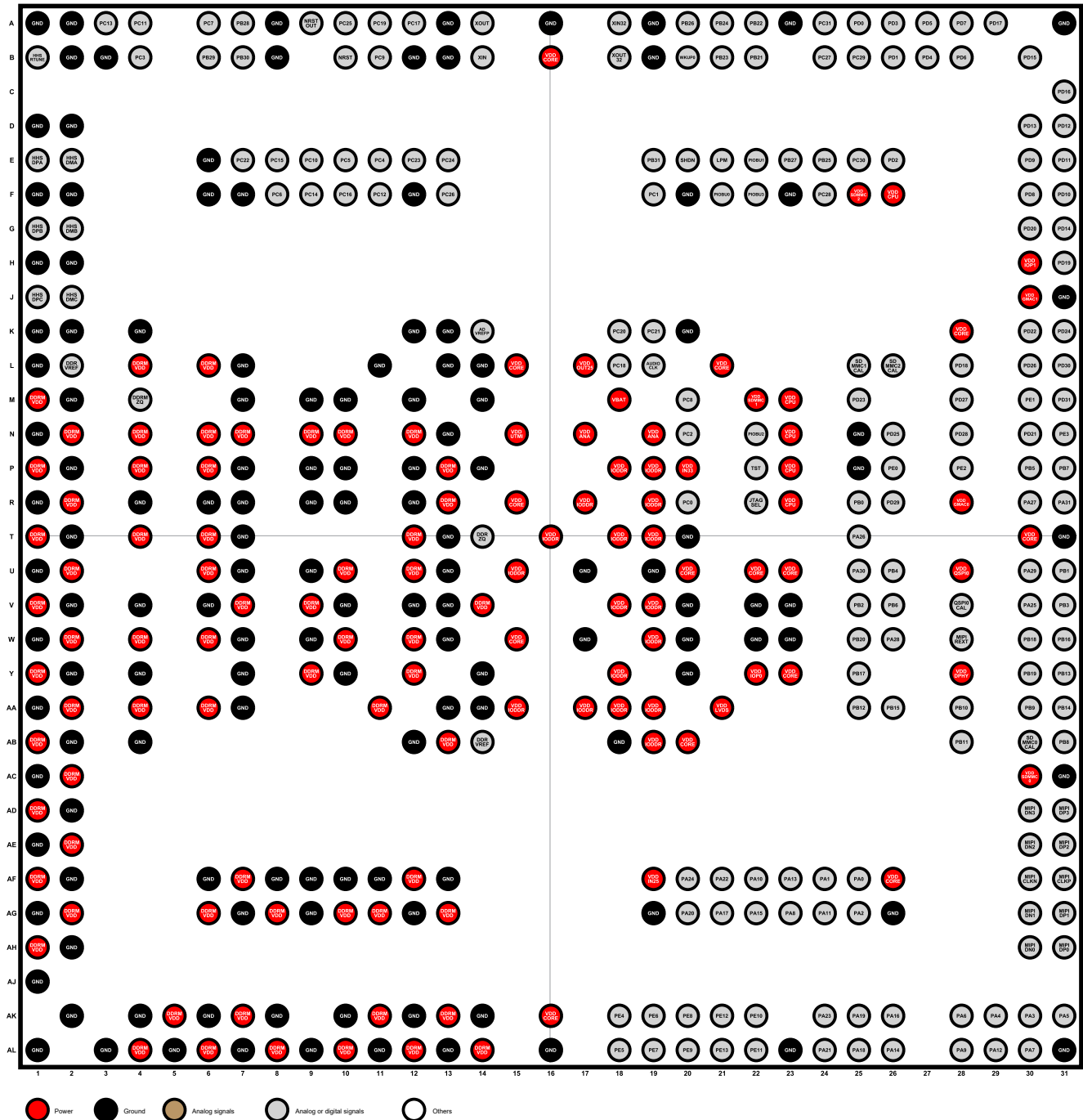
9.1. TFBGA427 Package

Figure 9-1. 427-Ball TFBGA Pinout



9.2. TFBGA442 Package

Figure 9-2. 442-Ball TFBGA Pinout



9.3. Pin Description

The device features several PIO controllers that multiplex the I/O lines of the peripheral set. [Table 9-1](#) defines how the I/O lines are multiplexed on the different PIO controllers. The "Reset State" column shows whether the PIO line resets in I/O mode or in Peripheral mode. If I/O is shown, the PIO line resets with the characteristics (input, output, pull-up or pull-down) indicated in this same column, so that the device is configured in a known state as soon as the reset is released. As a result, PIO_CFGR.FUNC resets to '0'. If a signal name is shown in the "Reset State" column, the PIO line is assigned to this function and PIO_CFGR.FUNC is not set to '0'. That is the case for pins controlling memories, in particular address lines, which require the pin to be driven as soon as the reset is released.

Table 9-1. Pin Description⁽¹⁾⁽²⁾

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
W21	AF25	VDDSDMMC0	HSIO	PA0	I/O	-	-	A	SDMMC0_CK	O	1	PIO, I, PU, ST
								B	FLEXCOM3_IO0	I/O	1	
								C	NWE/NWR0/NANDWE	O	1	
AA19	AF24	VDDSDMMC0	HSIO	PA1	I/O	-	-	A	SDMMC0_CMD	I/O	1	PIO, I, PU, ST
								B	FLEXCOM3_IO1	I/O	1	
								C	A21/NANDALE	O	1,2	
W19	AG25	VDDSDMMC0	HSIO	PA2	I/O	-	-	A	SDMMC0_RSTN	O	1	PIO, I, PU, ST
								B	FLEXCOM3_IO2	I/O	1	
								C	A22/NANDCLE	O	1,2	
Y19	AK30	VDDSDMMC0	HSIO	PA3	I/O	-	-	A	SDMMC0_DAT0	I/O	1	PIO, I, PU, ST
								B	FLEXCOM3_IO3	I/O	1	
								C	D0	I/O	1	
W20	AK29	VDDSDMMC0	HSIO	PA4	I/O	-	-	A	SDMMC0_DAT1	I/O	1	PIO, I, PU, ST
								B	FLEXCOM3_IO4	O	1	
								C	D1	I/O	1	
AA18	AK31	VDDSDMMC0	HSIO	PA5	I/O	-	-	A	SDMMC0_DAT4	I/O	1	PIO, I, PU, ST
								B	FLEXCOM2_IO0	I/O	3	
								C	D4	I/O	1	
								F	TCLK4	I	3	
U19	AK28	VDDSDMMC0	HSIO	PA6	I/O	-	-	A	SDMMC0_DAT5	I/O	1	PIO, I, PU, ST
								B	FLEXCOM2_IO1	I/O	3	
								C	D5	I/O	1	
								F	TIOB4	I/O	3	
AA17	AL30	VDDSDMMC0	HSIO	PA7	I/O	-	-	A	SDMMC0_DAT6	I/O	1	PIO, I, PU, ST
								B	FLEXCOM2_IO2	I/O	3	
								C	D6	I/O	1	
								F	TIOA4	I/O	3	
T18	AG23	VDDSDMMC0	HSIO	PA8	I/O	-	-	A	SDMMC0_DAT7	I/O	1	PIO, I, PU, ST
								B	FLEXCOM2_IO3	I/O	3	
								C	D7	I/O	1	
								F	TIOA5	I/O	3	
AA16	AL28	VDDSDMMC0	HSIO	PA9	I/O	-	-	A	SDMMC0_DAT2	I/O	1	PIO, I, PU, ST
								B	FLEXCOM0_IO2	I/O	1	
								C	D2	I/O	1	
								F	TIOB5	I/O	3	

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
W17	AF22	VDDSDMMC0	HSIO	PA10	I/O	-	-	A	SDMMC0_DAT3	I/O	1	PIO, I, PU, ST
								B	FLEXCOM0_IO3	I/O	1	
								C	D3	I/O	1	
								F	TCLK5	I	3	
V17	AG24	VDDSDMMC0	HSIO	PA11	I/O	-	-	A	SDMMC0_DS	I	1	PIO, I, PU, ST
								B	FLEXCOM0_IO4	O	1	
								C	NANDRDY	I	1,2	
								F	TIOB3	I/O	3	
Y17	AL29	VDDSDMMC0	HSIO	PA12	I/O	-	-	B	FLEXCOM0_IO0	I/O	1	PIO, I, PU, ST
								C	NRD/NANDOE	O	1,2	
								D	PCK0	O	1	
								E	EXT_IRQ0	I	1	
U17	AF23	VDDSDMMC0	HSIO	PA13	I/O	-	-	F	TIOA3	I/O	3	PIO, I, PU, ST
								B	FLEXCOM0_IO1	I/O	1	
								C	NCS0/NANDCS0	O	1,2	
								D	PCK1	O	1	
W13	AL26	VDDIOP0	GPIO	PA14	I/O	-	-	F	TCLK3	I	3	PIO, I, PU, ST
								A	FLEXCOM4_IO4	O	1	
								B	SDMMC0_WP	I	1	
								C	FLEXCOM3_IO0	I/O	4	
P15	AG22	VDDIOP0	GPIO	PA15	I/O	-	-	A	FLEXCOM4_IO3	I/O	1	PIO, I, PU, ST
								B	SDMMC0_1V8SEL	O	1	
								C	FLEXCOM3_IO1	I/O	4	
								A	FLEXCOM4_IO2	I/O	1	
V11	AK26	VDDIOP0	GPIO	PA16	I/O	-	-	B	SDMMC0_CD	I	1	PIO, I, PU, ST
								D	PCK2	O	1	
								E	EXT_IRQ1	I	1	
								A	FLEXCOM4_IO1	I/O	1	
R17	AG21	VDDIOP0	GPIO	PA17	I/O	-	-	A	FLEXCOM4_IO0	I/O	1	PIO, I, PU, ST
U11	AL25	VDDIOP0	GPIO	PA18	I/O	-	-	A	FLEXCOM4_IO0	I/O	1	PIO, I, PU, ST
V13	AK25	VDDIOP0	GPIO	PA19	I/O	-	-	A	TK0	I/O	1	PIO, I, PU, ST
								C	FLEXCOM4_IO5	O	1	
								D	PWML0	O	3	
								A	TD0	O	1	
T17	AG20	VDDIOP0	GPIO	PA20	I/O	-	-	B	FLEXCOM3_IO4	O	2	PIO, I, PU, ST
								C	FLEXCOM4_IO6	O	1	
								D	PWMH0	O	3	

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
U13	AL24	VDDIOP0	GPIO	PA21	I/O	-	-	A	TF0	I/O	1	PIO, I, PU, ST
								B	FLEXCOM3_IO3	I/O	2	
								D	PWML1	O	3	
R15	AF21	VDDIOP0	GPIO	PA22	I/O	-	-	A	RD0	I	1	PIO, I, PU, ST
								B	FLEXCOM3_IO2	I/O	2	
								C	PDMC0_DS1	I	1	
								D	PWMH1	O	3	
U15	AK24	VDDIOP0	GPIO	PA23	I/O	-	-	A	RK0	I/O	1	PIO, I, PU, ST
								B	FLEXCOM3_IO1	I/O	2	
								C	PDMC0_CLK	O	1	
								D	PWML2	O	3	
T13	AF20	VDDIOP0	GPIO	PA24	I/O	-	-	A	RF0	I/O	1	PIO, I, PU, ST
								B	FLEXCOM3_IO0	I/O	2	
								C	PDMC0_DS0	I	1	
								D	PWMH2	O	3	
R24	V30	VDDGMAC0	GPIO	PA25	I/O	-	-	A	G0_TXCTL/G0_TXEN	O	1,2	PIO, I, PU, ST
								B	FLEXCOM6_IO2	I/O	1	
R23	T25	VDDGMAC0	GPIO	PA26	I/O	-	-	A	G0_TX0	O	1,2	PIO, I, PU, ST
								B	FLEXCOM6_IO3	I/O	1	
N24	R30	VDDGMAC0	GPIO	PA27	I/O	-	-	A	G0_TX1	O	1,2	PIO, I, PU, ST
								B	FLEXCOM6_IO4	O	1	
L21	W26	VDDGMAC0	GPIO	PA28	I/O	-	-	A	G0_RXCTL/G0_CRSDV	I	1,2	PIO, I, PU, ST
								B	FLEXCOM6_IO0	I/O	1	
M25	U30	VDDGMAC0	GPIO	PA29	I/O	-	-	A	G0_RX0	I	1,2	PIO, I, PU, ST
								B	FLEXCOM6_IO1	I/O	1	
N23	U25	VDDGMAC0	GPIO	PA30	I/O	-	-	A	G0_RX1	I	1,2	PIO, I, PU, ST
								B	FLEXCOM8_IO0	I/O	1	
N25	R31	VDDGMAC0	GPIO	PA31	I/O	-	-	A	G0_MDC	O	1,2	PIO, I, PU, ST
								B	FLEXCOM8_IO1	I/O	1	
L20	R25	VDDGMAC0	GPIO	PB0	I/O	-	-	A	G0_MDIO	I/O	1,2	PIO, I, PU, ST
								B	FLEXCOM8_IO3	I/O	1	
P25	U31	VDDGMAC0	GPIO	PB1	I/O	-	-	A	G0_REFCK/G0_TXCK	I/O	2,1	PIO, I, PU, ST
								B	FLEXCOM8_IO2	I/O	1	
L23	V25	VDDGMAC0	GPIO	PB2	I/O	-	-	A	G0_RX2	I	1	PIO, I, PU, ST
								B	FLEXCOM8_IO4	O	1	
								C	G0_RXER	I	2	
								D	RK0	I/O	2	

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
R25	V31	VDDGMAC0	GPIO	PB3	I/O	-	-	A	G0_RXCK	I	1	PIO, I, PU, ST
								B	FLEXCOM10_IO2	I/O	2	
								D	TK0	I/O	2	
L24	U26	VDDGMAC0	GPIO	PB4	I/O	-	-	A	G0_TX2	O	1	PIO, I, PU, ST
								B	FLEXCOM10_IO3	I/O	2	
								D	TF0	I/O	2	
T25	P30	VDDGMAC0	GPIO	PB5	I/O	-	-	A	G0_TX3	O	1	PIO, I, PU, ST
								B	FLEXCOM10_IO4	O	2	
								D	TD0	O	2	
L19	V26	VDDGMAC0	GPIO	PB6	I/O	-	-	A	G0_RX3	I	1	PIO, I, PU, ST
								B	FLEXCOM10_IO0	I/O	2	
								D	RD0	I	2	
U25	P31	VDDGMAC0	GPIO	PB7	I/O	-	-	A	G0_TSUCOMP	O	1,2	PIO, I, PU, ST
								B	FLEXCOM10_IO1	I/O	2	
								C	ADTRG	I	1	
								D	RF0	I/O	2	
U23	AB31	VDDQSPI0	HSIO	PB8	I/O	-	-	A	QSPI0_IO3	I/O	1	PIO, I, PU, ST
								B	PCK3	O	1	
								D	FLEXCOM2_IO1	I/O	2	
V24	AA30	VDDQSPI0	HSIO	PB9	I/O	-	-	A	QSPI0_IO2	I/O	1	PIO, I, PU, ST
								D	FLEXCOM2_IO0	I/O	2	
								E	PWMEXTRG0	I	1	
N21	AA28	VDDQSPI0	HSIO	PB10	I/O	-	-	A	QSPI0_IO1	I/O	1	PIO, I, PU, ST
								D	FLEXCOM2_IO4	O	2	
								E	PWMEXTRG1	I	1	
U22	AB28	VDDQSPI0	HSIO	PB11	I/O	-	-	A	QSPI0_IO0	I/O	1	PIO, I, PU, ST
								D	FLEXCOM2_IO5	O	2	
								E	PWML3	O	1	
								F	TIOB3	I/O	2	
R20	AA25	VDDQSPI0	HSIO	PB12	I/O	-	-	A	QSPI0_CS	O	1	PIO, I, PU, ST
								D	FLEXCOM2_IO3	I/O	2	
								E	PWMFI1	I	1	
								F	TIOA3	I/O	2	
N20	Y31	VDDQSPI0	HSIO	PB13	I/O	-	-	A	QSPI0_SCK	O	1	PIO, I, PU, ST
								D	FLEXCOM2_IO2	I/O	2	
								E	PWMFI0	I	1	
								F	TCLK3	I	2	

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
R21	AA31	VDDQSPI0	HSIO	PB14	I/O	-	-	A	QSPI0_SCKN	O	1	PIO, I, PU, ST
								B	QSPI1_SCK	O	1	
								C	I2SMCC0_CK	I/O	3	
								D	FLEXCOM10_IO5	O	1	
								E	PWMH3	O	1	
								G	FLEXCOM2_IO1	I/O	4	
R22	AA26	VDDQSPI0	HSIO	PB15	I/O	-	-	A	QSPI0_IO4	I/O	1	PIO, I, PU, ST
								B	QSPI1_IO0	I/O	1	
								C	I2SMCC0_WS	I/O	3	
								D	FLEXCOM10_IO6	O	1	
								E	PWML0	O	1	
								F	TCLK4	I	2	
								G	FLEXCOM2_IO0	I/O	4	
U24	W31	VDDQSPI0	HSIO	PB16	I/O	-	-	A	QSPI0_IO5	I/O	1	PIO, I, PU, ST
								B	QSPI1_IO1	I/O	1	
								C	I2SMCC0_DIN0	I	3	
								D	FLEXCOM10_IO4	O	1	
								E	PWMH0	O	1	
								F	TIOB4	I/O	2	
M20	Y25	VDDQSPI0	HSIO	PB17	I/O	-	-	A	QSPI0_IO6	I/O	1	PIO, I, PU, ST
								B	QSPI1_IO2	I/O	1	
								C	I2SMCC0_DOUT0	O	3	
								D	FLEXCOM10_IO3	I/O	1	
								E	PWML1	O	1	
								F	TIOA4	I/O	2	
V25	W30	VDDQSPI0	HSIO	PB18	I/O	-	-	A	QSPI0_IO7	I/O	1	PIO, I, PU, ST
								B	QSPI1_IO3	I/O	1	
								C	I2SMCC0_MCK	O	3	
								D	FLEXCOM10_IO2	I/O	1	
								E	PWMH1	O	1	
								F	TIOA5	I/O	2	
N22	Y30	VDDQSPI0	HSIO	PB19	I/O	-	-	A	QSPI0_DQS	I	1	PIO, I, PU, ST
								B	EXT_IRQ1	I	2	
								C	PCK4	O	1	
								D	FLEXCOM10_IO1	I/O	1	
								E	PWML2	O	1	
								F	TIOB5	I/O	2	

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
M19	W25	VDDQSPI0	HSIO	PB20	I/O	-	-	A	QSPI0_INT	I	1	PIO, I, PU, ST
								B	QSPI1_CS	O	1	
								D	FLEXCOM10_IO0	I/O	1	
								E	PWMH2	O	1	
								F	TCLK5	I	2	
F17	B22	VDDSDMMC1	HSIO	PB21	I/O	-	-	A	SDMMC1_RSTN	O	1	PIO, I, PU, ST
								B	FLEXCOM6_IO4	O	2	
								C	TIOB2	I/O	2	
								D	ADTRG	I	2	
								E	EXT_IRQ0	I	2	
A18	A22	VDDSDMMC1	HSIO	PB22	I/O	-	-	A	SDMMC1_CMD	I/O	1	PIO, I, PU, ST
								B	FLEXCOM6_IO3	I/O	2	
								C	TCLK2	I	2	
D17	B21	VDDSDMMC1	HSIO	PB23	I/O	-	-	A	SDMMC1_CK	O	1	PIO, I, PU, ST
								B	FLEXCOM6_IO2	I/O	2	
								C	TIOA2	I/O	2	
A17	A21	VDDSDMMC1	HSIO	PB24	I/O	-	-	A	SDMMC1_DAT0	I/O	1	PIO, I, PU, ST
								B	FLEXCOM6_IO0	I/O	2	
C17	E24	VDDSDMMC1	HSIO	PB25	I/O	-	-	A	SDMMC1_DAT1	I/O	1	PIO, I, PU, ST
								B	FLEXCOM6_IO1	I/O	2	
								C	TIOB2	I/O	1	
B17	A20	VDDSDMMC1	HSIO	PB26	I/O	-	-	A	SDMMC1_DAT2	I/O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO0	I/O	3	
								C	TCLK2	I	1	
E17	E23	VDDSDMMC1	HSIO	PB27	I/O	-	-	A	SDMMC1_DAT3	I/O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO1	I/O	3	
								C	TIOA2	I/O	1	
A4	A7	VDDIN33	GPIO	PB28	I/O	AD12	I	A	SDMMC1_WP	I	1	PIO, I, PU, ST
								C	FLEXCOM1_IO0	I/O	3	
								E	D15	I/O	1,2	
E11	B6	VDDIN33	GPIO	PB29	I/O	AD13	I	A	SDMMC1_CD	I	1	PIO, I, PU, ST
								B	I2SMCC0_MCK	O	1	
								C	FLEXCOM1_IO1	I/O	3	
								E	D14	I/O	1,2	

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
B5	B7	VDDIN33	GPIO	PB30	I/O	AD14	I	A	SDMMC1_1V8SEL	O	1	PIO, I, PU, ST
								B	I2SMCC1_MCK	O	1	
								C	FLEXCOM1_IO2	I/O	3	
								D	TIOA1	I/O	1	
								E	NCS1/NANDCS1	O	1,2	
B12	E19	VDDIN33	GPIO	PB31	I/O	HHSА_СC1	I/O	A	PCK7	O	2	PIO, I, PU, ST
								B	I2SMCC1_DIN1	I	1	
								C	FLEXCOM1_IO3	I/O	3	
								D	TCLK1	I	1	
								E	NWE/NWR0/NANDWE	O	2	
F15	R20	VDDIN33	GPIO	PC0	I/O	HHSА_СC2	I/O	A	PCK6	O	2	PIO, I, PU, ST
								B	I2SMCC1_DIN2	I	1	
								C	FLEXCOM9_IO4	O	2	
								D	TIOB1	I/O	1	
								E	NWR1/NBS1	O	1,2	
B11	F19	VDDIN33	GPIO	PC1	I/O	HHSB_СC1	I/O	A	PCK5	O	1	PIO, I, PU, ST
								C	FLEXCOM9_IO2	I/O	2	
								E	SMCK	O	1,2	
E15	N20	VDDIN33	GPIO	PC2	I/O	HHSB_СC2	I/O	A	EXT_IRQ0	I	3	PIO, I, PU, ST
								C	FLEXCOM9_IO3	I/O	2	
								E	A11	O	1,2	
F11	B4	VDDIN33	GPIO	PC3	I/O	AD0	I	A	SPDIF_RX	I	2	PIO, I, PU, ST
								C	FLEXCOM9_IO0	I/O	2	
								D	FLEXCOM0_IO4	O	2	
								E	A10	O	1,2	
A3	E11	VDDIN33	GPIO	PC4	I/O	AD1	I	A	SPDIF_TX	O	2	PIO, I, PU, ST
								C	FLEXCOM9_IO1	I/O	2	
								D	FLEXCOM0_IO3	I/O	2	
								E	D0	I/O	2	
G13	E10	VDDIN33	GPIO	PC5	I/O	AD2	I	A	I3CC_SDASPUE	O	1	PIO, I, PU, ST
								B	I2SMCC1_DIN3	I	1	
								D	FLEXCOM0_IO2	I/O	2	
								E	D1	I/O	2	
J13	F8	VDDIN33	GPIO	PC6	I/O	AD3	I	A	I3CC_SCL	I/O	1	PIO, I, PU, ST
								D	FLEXCOM0_IO1	I/O	2	
								E	D4	I/O	2	

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
C9	A6	VDDIN33	GPIO	PC7	I/O	AD4	I	A	I3CC_SDA	I/O	1	PIO, I, PU, ST
								D	FLEXCOM0_IO0	I/O	2	
								E	D5	I/O	2	
D13	M20	VDDIN33	GPIO	PC8	I/O	ACC_INP1	I	A	I2SMCC0_DIN1	I	1	PIO, I, PU, ST
								B	PDMC0_DS1	I	2	
								C	I2SMCC1_DOUT1	O	1	
								D	FLEXCOM9_IO0	I/O	1	
								E	D6	I/O	2	
B7	B11	VDDIN33	GPIO	PC9	I/O	ACC_INN1	I	A	I2SMCC0_DIN2	I	1	PIO, I, PU, ST
								B	PDMC0_CLK	O	2	
								C	I2SMCC1_DOUT2	O	1	
								D	FLEXCOM9_IO1	I/O	1	
								E	D7	I/O	2	
H13	E9	VDDIN33	GPIO	PC10	I/O	AD5	I	A	I2SMCC0_DIN3	I	1	PIO, I, PU, ST
								B	PDMC0_DS0	I	2	
								C	I2SMCC1_DOUT3	O	1	
								D	FLEXCOM9_IO2	I/O	1	
								E	D2	I/O	2	
C5	A4	VDDIN33	GPIO	PC11	I/O	AD6	I	A	I2SMCC0_DOUT1	O	1	PIO, I, PU, ST
								B	PDMC1_DS0	I	1	
								D	FLEXCOM9_IO3	I/O	1	
								E	D3	I/O	2	
H11	F11	VDDIN33	GPIO	PC12	I/O	AD7	I	A	I2SMCC0_DOUT2	O	1	PIO, I, PU, ST
								B	PDMC1_CLK	O	1	
								D	FLEXCOM9_IO4	O	1	
								E	A9	O	1,2	
C7	A3	VDDIN33	GPIO	PC13	I/O	AD8	I	A	I2SMCC0_DOUT3	O	1	PIO, I, PD, ST
								B	PDMC1_DS1	I	1	
								E	A8	O	1,2	
J14	F9	VDDIN33	GPIO	PC14	I/O	AD9	I	A	I2SMCC1_DIN0	I	1	PIO, I, PD, ST
								B	SPDIF_RX	I	3	
								C	FLEXCOM1_IO0	I/O	2	
								E	A7	O	1,2	
D7	E8	VDDIN33	GPIO	PC15	I/O	AD10	I	A	I2SMCC1_WS	I/O	1	PIO, I, PD, ST
								B	PDMC1_DS1	I	2	
								C	FLEXCOM1_IO1	I/O	2	
								E	A6	O	1,2	

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
J11	F10	VDDIN33	GPIO	PC16	I/O	AD11	I	A	I2SMCC1_CK	I/O	1	PIO, I, PD, ST
								B	PDMC1_CLK	O	2	
								C	FLEXCOM1_IO2	I/O	2	
								D	TIOA1	I/O	2	
								E	A5	O	1,2	
A8	A12	VDDIN33	GPIO	PC17	I/O	ACC_INP0	I	A	I2SMCC1_DOUT0	O	1	PIO, I, PU, ST
								B	PDMC1_DS0	I	2	
								C	FLEXCOM1_IO3	I/O	2	
								D	TCLK1	I	2	
								E	A4	O	1,2	
E13	L18	VDDIN33	GPIO	PC18	I/O	ACC_INN2	I	A	I2SMCC0_DIN0	I	1	PIO, I, PU, ST
								B	SPDIF_TX	O	3	
								C	FLEXCOM1_IO4	O	2	
								D	TIOB1	I/O	2	
								E	A3	O	1,2	
A7	A11	VDDIN33	GPIO	PC19	I/O	ACC_INP2	I	A	I2SMCC0_WS	I/O	1	PIO, I, PU, ST
								B	PCK6	O	1	
								E	A2	O	1,2	
A6	K18	VDDIN33	GPIO	PC20	I/O	ACC_INN3	I	A	I2SMCC0_DOUT0	O	1	PIO, I, PU, ST
								E	A1	O	1,2	
								A	I2SMCC0_CK	I/O	1	
G15	K19	VDDIN33	GPIO	PC21	I/O	ACC_INP3	I	B	PCK7	O	1	PIO, I, PU, ST
								E	A0/NBS0	O	1,2	
								A	NTRST	I	1	NTRST, PU, ST
D9	E7	VDDIN33	GPIO	PC22	I/O	AD15	I	E	NWAIT	I	1,2	
								A	TCK_SWCLK	I	1	TCK_SWCLK, ST
D11	E13	VDDIN33	GPIO	PC24	I/O	-	-	A	TMS_SWDIO	I/O	1	TMS_SWDIO, PU, ST
B6	A10	VDDIN33	GPIO	PC25	I/O	-	-	A	TDI	I	1	TDI, PU, ST
F13	F13	VDDIN33	GPIO	PC26	I/O	-	-	A	TDO	O	1	TDO, ST
								E	A12	O	1,2	
								A	SDMMC2_CMD	I/O	1	PIO, I, PU, ST
A19	B24	VDDSDMMC2	HSIO	PC27	I/O	-	-	B	FLEXCOM8_IO0	I/O	2	
								D	TD1	O	2	
								E	D8	I/O	1,2	

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
B19	F24	VDDSDMMC2	HSIO	PC28	I/O	-	-	A	SDMMC2_CK	O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO1	I/O	2	
								D	TF1	I/O	2	
								E	D9	I/O	1,2	
A20	B25	VDDSDMMC2	HSIO	PC29	I/O	-	-	A	SDMMC2_DAT0	I/O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO2	I/O	2	
								D	TK1	I/O	2	
								E	D10	I/O	1,2	
								F	TCLK0	I	1	
C19	E25	VDDSDMMC2	HSIO	PC30	I/O	-	-	A	SDMMC2_DAT1	I/O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO3	I/O	2	
								D	RD1	I	2	
								E	D11	I/O	1,2	
								F	TIOA0	I/O	1	
B20	A24	VDDSDMMC2	HSIO	PC31	I/O	-	-	A	SDMMC2_DAT2	I/O	1	PIO, I, PU, ST
								B	FLEXCOM8_IO4	O	2	
								C	PCK0	O	2	
								D	RK1	I/O	2	
								E	D12	I/O	1,2	
								F	TIOB0	I/O	1	
D20	A25	VDDSDMMC2	HSIO	PD0	I/O	-	-	A	SDMMC2_DAT3	I/O	1	PIO, I, PU, ST
								C	PCK1	O	2	
								D	RF1	I/O	2	
								E	D13	I/O	1,2	
A21	B26	VDDIOP1	GPIO	PD1	I/O	-	-	A	SDMMC2_WP	I	1	PIO, I, PU, ST
								B	FLEXCOM1_IO5	O	1	
								C	LCDC_HSYNC	O	2	
								D	FLEXCOM3_IO0	I/O	3	
D19	E26	VDDIOP1	GPIO	PD2	I/O	-	-	A	SDMMC2_CD	I	1	PIO, I, PU, ST
								B	FLEXCOM1_IO6	O	1	
								C	LCDC_VSYNC	O	2	
								D	FLEXCOM3_IO1	I/O	3	
B21	A26	VDDIOP1	GPIO	PD3	I/O	-	-	A	SDMMC2_1V8SEL	O	1	PIO, I, PU, ST
								B	FLEXCOM1_IO4	O	1	
								C	TIOA0	I/O	2	
								D	FLEXCOM3_IO2	I/O	3	
								E	EXT_IRQ1	I	3	

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
G20	B27	VDDIOP1	GPIO	PD4	I/O	-	-	A	LCDC_HSYNC	O	1	PIO, I, PU, ST
								B	FLEXCOM1_IO2	I/O	1	
								C	TIOB0	I/O	2	
								D	FLEXCOM7_IO1	I/O	3	
C21	A27	VDDIOP1	GPIO	PD5	I/O	-	-	A	LCDC_VSYNC	O	1	PIO, I, PU, ST
								B	FLEXCOM1_IO3	I/O	1	
								C	TCLK0	I	2	
								D	FLEXCOM7_IO0	I/O	3	
E21	B28	VDDIOP1	GPIO	PD6	I/O	-	-	A	LCDC_PWM	O	1,2	PIO, I, PU, ST
								B	FLEXCOM1_IO1	I/O	1	
								D	FLEXCOM7_IO2	I/O	3	
B22	A28	VDDIOP1	GPIO	PD7	I/O	-	-	A	LCDC_DISP	O	1,2	PIO, I, PU, ST
								B	FLEXCOM1_IO0	I/O	1	
								D	FLEXCOM7_IO3	I/O	3	
G21	F30	VDDIOP1	GPIO	PD8	I/O	-	-	A	CANTX0	O	1	PIO, I, PU, ST
								B	FLEXCOM7_IO0	I/O	1	
A22	E30	VDDIOP1	GPIO	PD9	I/O	-	-	A	CANRX0	I	1	PIO, I, PU, ST
								B	FLEXCOM7_IO1	I/O	1	
E22	F31	VDDIOP1	GPIO	PD10	I/O	-	-	A	CANTX1	O	1	PIO, I, PU, ST
								B	FLEXCOM7_IO2	I/O	1	
								C	TIOA1	I/O	3	
A23	E31	VDDIOP1	GPIO	PD11	I/O	-	-	A	CANRX1	I	1	PIO, I, PU, ST
								B	FLEXCOM7_IO3	I/O	1	
								C	TCLK1	I	3	
B25	D31	VDDIOP1	GPIO	PD12	I/O	-	-	A	CANTX2	O	1	PIO, I, PU, ST
								B	FLEXCOM7_IO4	O	1	
								C	TIOB1	I/O	3	
								D	PCK2	O	2	
								E	FLEXCOM3_IO3	I/O	3	
C23	D30	VDDIOP1	GPIO	PD13	I/O	-	-	A	CANRX2	I	1	PIO, I, PU, ST
								B	FLEXCOM5_IO4	O	1	
								C	TIOA2	I/O	3	
								D	PCK3	O	2	
C25	G31	VDDIOP1	GPIO	PD14	I/O	-	-	A	CANTX3	O	1	PIO, I, PU, ST
								B	FLEXCOM5_IO2	I/O	1	
								C	TIOB2	I/O	3	

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
C24	B30	VDDIOP1	GPIO	PD15	I/O	-	-	A	CANRX3	I	1	PIO, I, PU, ST
								B	FLEXCOM5_IO3	I/O	1	
								C	TCLK2	I	3	
D25	C31	VDDIOP1	GPIO	PD16	I/O	-	-	A	CANTX4	O	1	PIO, I, PU, ST
								B	FLEXCOM5_IO0	I/O	1	
E23	A29	VDDIOP1	GPIO	PD17	I/O	-	-	A	CANRX4	I	1	PIO, I, PU, ST
								B	FLEXCOM5_IO1	I/O	1	
G22	L28	VDDGMAC1	GPIO	PD18	I/O	-	-	B	FLEXCOM6_IO0	I/O	4	PIO, I, PU, ST
								C	CANTX1	O	2	
								D	PCK4	O	2	
E24	H31	VDDGMAC1	GPIO	PD19	I/O	-	-	B	FLEXCOM6_IO1	I/O	4	PIO, I, PU, ST
								C	CANRX1	I	2	
								D	PCK2	O	3	
J18	G30	VDDGMAC1	GPIO	PD20	I/O	-	-	B	FLEXCOM6_IO2	I/O	4	PIO, I, PU, ST
								C	I2SMCC1_MCK	O	2	
								D	PCK3	O	3	
K19	N30	VDDGMAC1	GPIO	PD21	I/O	-	-	A	G1_TXCTL/G1_TXEN	O	1,2	PIO, I, PU, ST
								B	FLEXCOM6_IO2	I/O	3	
								C	TK1	I/O	1	
E25	K30	VDDGMAC1	GPIO	PD22	I/O	-	-	A	G1_TX0	O	1,2	PIO, I, PU, ST
								B	FLEXCOM6_IO3	I/O	3,4	
								C	TF1	I/O	1	
G23	M25	VDDGMAC1	GPIO	PD23	I/O	-	-	A	G1_TX1	O	1,2	PIO, I, PU, ST
								B	FLEXCOM6_IO4	O	3,4	
								C	TD1	O	1	
F25	K31	VDDGMAC1	GPIO	PD24	I/O	-	-	A	G1_RXCTL/G1_CRSDV	I	1,2	PIO, I, PU, ST
								B	FLEXCOM6_IO0	I/O	3	
								C	RD1	I	1	
								E	PDMC0_DS1	I	3	
J20	N26	VDDGMAC1	GPIO	PD25	I/O	-	-	A	G1_MDC	O	1,2	PIO, I, PU, ST
								B	FLEXCOM6_IO1	I/O	3	
								C	RK1	I/O	1	
								E	PDMC0_CLK	O	3	
G25	L30	VDDGMAC1	GPIO	PD26	I/O	-	-	A	G1_MDIO	I/O	1,2	PIO, I, PU, ST
								B	FLEXCOM7_IO4	O	2	
								C	RF1	I/O	1	
								D	I2SMCC1_DIN2	I	2	
								E	PDMC0_DS0	I	3	

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
J21	M28	VDDGMAC1	GPIO	PD27	I/O	-	-	A	G1_RX0	I	1,2	PIO, I, PU, ST
								B	FLEXCOM7_IO0	I/O	2	
								C	SPDIF_RX	I	1	
								D	I2SMCC1_DIN3	I	2	
G24	N28	VDDGMAC1	GPIO	PD28	I/O	-	-	A	G1_RX1	I	1,2	PIO, I, PU, ST
								B	FLEXCOM7_IO1	I/O	2	
								C	SPDIF_TX	O	1	
								D	I2SMCC1_DIN1	I	2	
J24	R26	VDDGMAC1	GPIO	PD29	I/O	-	-	A	G1_REFCK/G1_TXCK	I/O	2,1	PIO, I, PU, ST
								B	FLEXCOM7_IO2	I/O	2	
								C	I2SMCC1_DOUT3	O	2	
H25	L31	VDDGMAC1	GPIO	PD30	I/O	-	-	A	G1_RX2	I	1	PIO, I, PU, ST
								B	FLEXCOM7_IO3	I/O	2	
								C	I2SMCC1_DOUT1	O	2	
								D	PDMC1_DS1	I	3	
								E	G1_RXER	I	2	
J25	M31	VDDGMAC1	GPIO	PD31	I/O	-	-	A	G1_RX3	I	1	PIO, I, PU, ST
								B	FLEXCOM5_IO4	O	2	
								C	I2SMCC1_DOUT2	O	2	
								D	PDMC1_DS0	I	3	
J23	P26	VDDGMAC1	GPIO	PE0	I/O	-	-	A	G1_TX2	O	1	PIO, I, PU, ST
								B	FLEXCOM5_IO2	I/O	2	
								C	I2SMCC1_DIN0	I	2	
								D	PDMC1_CLK	O	3	
K25	M30	VDDGMAC1	GPIO	PE1	I/O	-	-	A	G1_TX3	O	1	PIO, I, PU, ST
								B	FLEXCOM5_IO3	I/O	2	
								C	I2SMCC1_WS	I/O	2	
								D	PDMC0_DS1	I	4	
J22	P28	VDDGMAC1	GPIO	PE2	I/O	-	-	A	G1_RXCK	I	1	PIO, I, PU, ST
								B	FLEXCOM5_IO1	I/O	2	
								C	I2SMCC1_CK	I/O	2	
								D	PDMC0_CLK	O	4	
L25	N31	VDDGMAC1	GPIO	PE3	I/O	-	-	A	G1_TSUCOMP	O	1,2	PIO, I, PU, ST
								B	FLEXCOM5_IO0	I/O	2	
								C	I2SMCC1_DOUT0	O	2	
								D	PDMC0_DS0	I	4	

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
Y11	AK18	VDDLVD	GPIO	PE4	I/O	LVDS_A0M	O	A	LCDC_DAT0	O	1,2	PIO, I, PU, ST
								B	FLEXCOM2_IO2	I/O	1	
								C	PWML0	O	2	
								D	TIOA3	I/O	1	
								E	I2SMCC0_DIN1	I	2	
AA11	AL18	VDDLVD	GPIO	PE5	I/O	LVDS_A0P	O	A	LCDC_DAT1	O	1,2	PIO, I, PU, ST
								B	FLEXCOM2_IO3	I/O	1	
								C	PWMH0	O	2	
								D	TIOB3	I/O	1	
								E	I2SMCC0_DIN2	I	2	
Y12	AK19	VDDLVD	GPIO	PE6	I/O	LVDS_A1M	O	A	LCDC_DAT2	O	1,2	PIO, I, PU, ST
								B	FLEXCOM2_IO4	O	1	
								C	PWML1	O	2	
								D	TCLK3	I	1	
								E	I2SMCC0_DIN3	I	2	
AA12	AL19	VDDLVD	GPIO	PE7	I/O	LVDS_A1P	O	A	LCDC_DAT3	O	1,2	PIO, I, PU, ST
								B	FLEXCOM2_IO5	O	1	
								C	PWMH1	O	2	
								D	TIOA4	I/O	1	
								E	I2SMCC0_DOUT1	O	2	
Y13	AK20	VDDLVD	GPIO	PE8	I/O	LVDS_A2M	O	A	LCDC_DAT4	O	1,2	PIO, I, PU, ST
								B	FLEXCOM2_IO0	I/O	1	
								C	PWML2	O	2	
								D	TIOB4	I/O	1	
								E	I2SMCC0_CK	I/O	2	
AA13	AL20	VDDLVD	GPIO	PE9	I/O	LVDS_A2P	O	A	LCDC_DAT5	O	1,2	PIO, I, PU, ST
								B	FLEXCOM2_IO1	I/O	1	
								C	PWMH2	O	2	
								D	TCLK4	I	1	
								E	I2SMCC0_WS	I/O	2	
Y15	AK22	VDDLVD	GPIO	PE10	I/O	LVDS_A3M	O	A	LCDC_DAT6	O	1,2	PIO, I, PU, ST
								B	FLEXCOM2_IO6	O	1	
								C	PWML3	O	2	
								D	TIOA5	I/O	1	
								E	I2SMCC0_DOUT2	O	2	

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
AA15	AL22	VDDLVD5	GPIO	PE11	I/O	LVDS_A3P	O	A	LCDC_DAT7	O	1,2	PIO, I, PU, ST
								C	PWMH3	O	2	
								D	TIOB5	I/O	1	
								E	I2SMCC0_DOUT3	O	2	
Y14	AK21	VDDLVD5	GPIO	PE12	I/O	LVDS_CLK1M	O	A	LCDC_DEN	O	1,2	PIO, I, PU, ST
								B	PCK3	O	4	
								C	PWMEXTRG0	I	2	
								D	TCLK5	I	1	
								E	I2SMCC0_DIN0	I	2	
AA14	AL21	VDDLVD5	GPIO	PE13	I/O	LVDS_CLK1P	O	A	LCDC_PCK	O	1,2	PIO, I, PU, ST
								B	PCK4	O	3	
								C	PWMEXTRG1	I	2	
								E	I2SMCC0_DOUT0	O	2	
B4	K14	ADVREFP	Analog input	ADVREFP	I	-	-	-	-	-	-	-
F14	P20	VDDIN33	power	VDDIN33	I	-	-	-	-	-	-	-
H16	-	GNDIN33	ground	GNDIN33	I	-	-	-	-	-	-	-
F16	-	GNDANA	ground	GNDANA	I	-	-	-	-	-	-	-
E10	-	GNDANA	ground	GNDANA	I	-	-	-	-	-	-	-
E12	-	GNDANA	ground	GNDANA	I	-	-	-	-	-	-	-
N16	-	GNDANA	ground	GNDANA	I	-	-	-	-	-	-	-
Y18	Y22	VDDIOP0	power	VDDIOP0	I	-	-	-	-	-	-	-
K24	R28	VDDGMAC0	power	VDDGMAC0	I	-	-	-	-	-	-	-
F21	H30	VDDIOP1	power	VDDIOP1	I	-	-	-	-	-	-	-
F24	J30	VDDGMAC1	power	VDDGMAC1	I	-	-	-	-	-	-	-
A1	A1	GND	ground	GND	I	-	-	-	-	-	-	-
A25	A2	GND	ground	GND	I	-	-	-	-	-	-	-
AA1	A8	GND	ground	GND	I	-	-	-	-	-	-	-
AA4	A13	GND	ground	GND	I	-	-	-	-	-	-	-
AA6	A16	GND	ground	GND	I	-	-	-	-	-	-	-
AA8	A19	GND	ground	GND	I	-	-	-	-	-	-	-
AA10	A23	GND	ground	GND	I	-	-	-	-	-	-	-
AA25	A31	GND	ground	GND	I	-	-	-	-	-	-	-
B16	B2	GND	ground	GND	I	-	-	-	-	-	-	-
B24	B3	GND	ground	GND	I	-	-	-	-	-	-	-
D4	B8	GND	ground	GND	I	-	-	-	-	-	-	-
D24	B13	GND	ground	GND	I	-	-	-	-	-	-	-
E16	N17	VDDANA	power	VDDANA	I	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
F10	N19	VDDANA	power	VDDANA	I	-	-	-	-	-	-	-
V15	AF19	VDDIN25	power	VDDIN25	O	-	-	-	-	-	-	-
F12	L17	VDDOUT25	power	VDDOUT25	I	-	-	-	-	-	-	-
G11	B16	VDDCORE	power	VDDCORE	I	-	-	-	-	-	-	-
H12	K28	VDDCORE	power	VDDCORE	I	-	-	-	-	-	-	-
J17	L15	VDDCORE	power	VDDCORE	I	-	-	-	-	-	-	-
K17	L21	VDDCORE	power	VDDCORE	I	-	-	-	-	-	-	-
K20	R15	VDDCORE	power	VDDCORE	I	-	-	-	-	-	-	-
L13	T30	VDDCORE	power	VDDCORE	I	-	-	-	-	-	-	-
M12	U20	VDDCORE	power	VDDCORE	I	-	-	-	-	-	-	-
N17	U22	VDDCORE	power	VDDCORE	I	-	-	-	-	-	-	-
N19	U23	VDDCORE	power	VDDCORE	I	-	-	-	-	-	-	-
P21	W15	VDDCORE	power	VDDCORE	I	-	-	-	-	-	-	-
T14	Y23	VDDCORE	power	VDDCORE	I	-	-	-	-	-	-	-
T24	AB20	VDDCORE	power	VDDCORE	I	-	-	-	-	-	-	-
V19	AF26	VDDCORE	power	VDDCORE	I	-	-	-	-	-	-	-
Y16	AK16	VDDCORE	power	VDDCORE	I	-	-	-	-	-	-	-
G19	F26	VDDCPU	power	VDDCPU	I	-	-	-	-	-	-	-
H20	M23	VDDCPU	power	VDDCPU	I	-	-	-	-	-	-	-
H24	N23	VDDCPU	power	VDDCPU	I	-	-	-	-	-	-	-
J19	P23	VDDCPU	power	VDDCPU	I	-	-	-	-	-	-	-
K18	R23	VDDCPU	power	VDDCPU	I	-	-	-	-	-	-	-
E2	B19	GND	ground	GND	I	-	-	-	-	-	-	-
E5	D1	GND	ground	GND	I	-	-	-	-	-	-	-
E8	D2	GND	ground	GND	I	-	-	-	-	-	-	-
E14	E6	GND	ground	GND	I	-	-	-	-	-	-	-
E18	F1	GND	ground	GND	I	-	-	-	-	-	-	-
F1	F2	GND	ground	GND	I	-	-	-	-	-	-	-
F9	F6	GND	ground	GND	I	-	-	-	-	-	-	-
F20	F7	GND	ground	GND	I	-	-	-	-	-	-	-
G2	F12	GND	ground	GND	I	-	-	-	-	-	-	-
G3	H1	GND	ground	GND	I	-	-	-	-	-	-	-
G5	H2	GND	ground	GND	I	-	-	-	-	-	-	-
G7	J31	GND	ground	GND	I	-	-	-	-	-	-	-
G9	K1	GND	ground	GND	I	-	-	-	-	-	-	-
H1	K2	GND	ground	GND	I	-	-	-	-	-	-	-
H5	K4	GND	ground	GND	I	-	-	-	-	-	-	-
H8	K12	GND	ground	GND	I	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
H10	K13	GND	ground	GND	I	-	-	-	-	-	-	-
H18	L1	GND	ground	GND	I	-	-	-	-	-	-	-
P16	L7	GND	ground	GND	I	-	-	-	-	-	-	-
T16	AA21	VDDLVS	power	VDDLVS	I	-	-	-	-	-	-	-
H14	-	GNDUTMI	ground	GNDUTMI	I	-	-	-	-	-	-	-
J15	N15	VDDUTMI	power	VDDUTMI	I	-	-	-	-	-	-	-
B1	E1	VDDUTMI	-	HHSDPA	I/O	-	-	-	-	-	-	-
B2	E2	VDDUTMI	-	HHSDMA	I/O	-	-	-	-	-	-	-
C1	G1	VDDUTMI	-	HHSDPB	I/O	-	-	-	-	-	-	-
C2	G2	VDDUTMI	-	HHSDMB	I/O	-	-	-	-	-	-	-
D1	J1	VDDUTMI	-	HHSDPC	I/O	-	-	-	-	-	-	-
D2	J2	VDDUTMI	-	HHSDMC	I/O	-	-	-	-	-	-	-
C3	B1	VDDUTMI	Analog input	HHSRTUNE	I	-	-	-	-	-	-	-
K12	P18	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
K14	P19	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
L9	R17	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
L10	R19	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
L11	T16	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
L15	T18	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
M9	T19	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
M10	U15	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
M14	V18	VDDIODDR	power	VDDIODDR	I	2	-	-	-	-	-	-
M16	V19	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
N11	W19	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
N13	Y18	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
N15	AA15	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
P10	AA17	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
P12	AA18	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
H21	L11	GND	ground	GND	I	-	-	-	-	-	-	-
J1	L13	GND	ground	GND	I	-	-	-	-	-	-	-
J3	L14	GND	ground	GND	I	-	-	-	-	-	-	-
J5	M2	GND	ground	GND	I	-	-	-	-	-	-	-
J7	M7	GND	ground	GND	I	-	-	-	-	-	-	-
J9	M9	GND	ground	GND	I	-	-	-	-	-	-	-
J10	M10	GND	ground	GND	I	-	-	-	-	-	-	-
K2	M12	GND	ground	GND	I	-	-	-	-	-	-	-
K8	M14	GND	ground	GND	I	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
K10	N1	GND	ground	GND	I	-	-	-	-	-	-	-
K11	N13	GND	ground	GND	I	-	-	-	-	-	-	-
K13	N25	GND	ground	GND	I	-	-	-	-	-	-	-
K15	P2	GND	ground	GND	I	-	-	-	-	-	-	-
K21	P7	GND	ground	GND	I	-	-	-	-	-	-	-
L1	P9	GND	ground	GND	I	-	-	-	-	-	-	-
E4	L2	VDDIODDR	-	DDR_VREF	I	-	-	-	-	-	-	-
J12	T14	VDDIODDR	Analog input	DDR_ZQ	O	-	-	-	-	-	-	-
P20	Y28	VDDDDPHY	power	VDDDDPHY	I	-	-	-	-	-	-	-
M21	-	GNDDDPHY	ground	GNDDDPHY	I	-	-	-	-	-	-	-
Y22	AF30	VDDDDPHY	-	MIPI_CLKN	O	-	-	-	-	-	-	-
AA22	AF31	VDDDDPHY	-	MIPI_CLKP	O	-	-	-	-	-	-	-
Y20	AH30	VDDDDPHY	-	MIPI_DN0	O	-	-	-	-	-	-	-
AA20	AH31	VDDDDPHY	-	MIPI_DP0	O	-	-	-	-	-	-	-
Y21	AG30	VDDDDPHY	-	MIPI_DN1	O	-	-	-	-	-	-	-
AA21	AG31	VDDDDPHY	-	MIPI_DP1	O	-	-	-	-	-	-	-
Y24	AE30	VDDDDPHY	-	MIPI_DN2	O	-	-	-	-	-	-	-
Y25	AE31	VDDDDPHY	-	MIPI_DP2	O	-	-	-	-	-	-	-
W24	AD30	VDDDDPHY	-	MIPI_DN3	O	-	-	-	-	-	-	-
W25	AD31	VDDDDPHY	-	MIPI_DP3	O	-	-	-	-	-	-	-
AA23	W28	VDDDDPHY	-	MIPI_REXT	I	-	-	-	-	-	-	-
U21	AC30	VDDSDMMC0	power	VDDSDMMC0	I	-	-	-	-	-	-	-
B18	M22	VDDSDMMC1	power	VDDSDMMC1	I	-	-	-	-	-	-	-
T20	AB30	VDDSDMMC0	Analog input	SDMMC0_CAL	I	-	-	-	-	-	-	-
F18	L25	VDDSDMMC1	Analog input	SDMMC1_CAL	I	-	-	-	-	-	-	-
D22	F25	VDDSDMMC2	power	VDDSDMMC2	I	-	-	-	-	-	-	-
E19	L26	VDDSDMMC2	Analog input	SDMMC2_CAL	I	-	-	-	-	-	-	-
J16	-	GNDBAT	ground	GNDBAT	I	-	-	-	-	-	-	-
K16	M18	VBAT	power	VBAT	I	-	-	-	-	-	-	-
G17	F21	VBAT	PIOBU	PIOBU0	I/O	-	-	-	-	-	-	PU ⁽⁶⁾
B14	E22	VBAT	PIOBU	PIOBU1	I/O	-	-	-	-	-	-	PU ⁽⁶⁾
C15	N22	VBAT	PIOBU	PIOBU2	I/O	-	-	-	-	-	-	PU ⁽⁶⁾
A13	F22	VBAT	PIOBU	PIOBU3	I/O	-	-	-	-	-	-	PD ⁽⁶⁾
B10	B14	VDDIN33	-	XIN	I	-	-	-	-	-	-	-
A10	A14	VDDIN33	-	XOUT	O	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
A15	A18	VBAT	-	XIN32	I	-	-	-	-	-	-	-
B15	B18	VBAT	-	XOUT32	O	-	-	-	-	-	-	-
D15	P22	VBAT	-	TST	I	-	-	-	-	-	-	PD
C13	R22	VBAT	-	JTAGSEL	I	-	-	-	-	-	-	PD
A16	B20	VBAT	-	WKUP0	I	-	-	-	-	-	-	-
B13	E20	VBAT	-	SHDN	O	-	-	-	-	-	-	-
B9	B10	VDDIN33	-	NRST	I	-	-	-	-	-	-	PU
B8	A9	VDDIN33	-	NRST_OUT	O	-	-	-	-	-	-	OD ⁽⁷⁾
C11	L19	VDDIN33	GPIO	AUDIOCLK	O	-	-	-	-	-	-	-
A12	E21	VBAT	-	LPM	O	-	-	-	-	-	-	-
M24	U28	VDDQSPI0	power	VDDQSPI0	I	-	-	-	-	-	-	-
L22	V28	VDDQSPI0	Analog input	QSPI0_CAL	I	-	-	-	-	-	-	-
A14	F23	GND	ground	GND	I	-	-	-	-	-	-	-
H15	K20	GND	ground	GND	I	-	-	-	-	-	-	-
A11	F20	GND	ground	GND	I	-	-	-	-	-	-	-
A9	B12	GND	ground	GND	I	-	-	-	-	-	-	-
AA3	L4	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
AA5	L6	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
AA7	M1	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
AA9	N2	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
E1	N4	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
E3	N6	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
E7	N7	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
E9	N9	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
F2	N10	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
F5	N12	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
F6	P1	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
F8	P4	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
G1	P6	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
G4	M4	DDRM_VDD	power	DDRM_ZQ	I	-	-	-	-	-	-	-
G6	P13	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
H2	R2	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
H6	R13	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
J2	T1	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
J4	T4	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
J6	T6	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
K1	T12	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
K5	U2	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
K6	U6	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
K7	U10	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
K9	U12	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
L2	V1	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
L3	P10	GND	ground	GND	I	-	-	-	-	-	-	-
L4	V7	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
L5	P12	GND	ground	GND	I	-	-	-	-	-	-	-
L6	V9	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
L7	P14	GND	ground	GND	I	-	-	-	-	-	-	-
L8	P25	GND	ground	GND	I	-	-	-	-	-	-	-
L12	R1	GND	ground	GND	I	-	-	-	-	-	-	-
L14	R4	GND	ground	GND	I	-	-	-	-	-	-	-
L16	R6	GND	ground	GND	I	-	-	-	-	-	-	-
L17	R7	GND	ground	GND	I	-	-	-	-	-	-	-
L18	R9	GND	ground	GND	I	-	-	-	-	-	-	-
M1	V14	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
M2	R10	GND	ground	GND	I	-	-	-	-	-	-	-
M5	R12	GND	ground	GND	I	-	-	-	-	-	-	-
M6	T2	GND	ground	GND	I	-	-	-	-	-	-	-
M7	W2	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
M8	T7	GND	ground	GND	I	-	-	-	-	-	-	-
M11	T13	GND	ground	GND	I	-	-	-	-	-	-	-
M13	T20	GND	ground	GND	I	-	-	-	-	-	-	-
M15	T31	GND	ground	GND	I	-	-	-	-	-	-	-
M17	U1	GND	ground	GND	I	-	-	-	-	-	-	-
M18	U7	GND	ground	GND	I	-	-	-	-	-	-	-
N1	U9	GND	ground	GND	I	-	-	-	-	-	-	-
N2	U13	GND	ground	GND	I	-	-	-	-	-	-	-
N3	W4	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
N4	U17	GND	ground	GND	I	-	-	-	-	-	-	-
N5	W6	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
N6	W10	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
N7	U19	GND	ground	GND	I	-	-	-	-	-	-	-
N8	W12	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
N9	V2	GND	ground	GND	I	-	-	-	-	-	-	-
N10	V4	GND	ground	GND	I	-	-	-	-	-	-	-
N12	V6	GND	ground	GND	I	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
N14	V10	GND	ground	GND	I	-	-	-	-	-	-	-
N18	V12	GND	ground	GND	I	-	-	-	-	-	-	-
P1	Y1	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
P2	V13	GND	ground	GND	I	-	-	-	-	-	-	-
P5	V20	GND	ground	GND	I	-	-	-	-	-	-	-
P6	Y9	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
P8	V22	GND	ground	GND	I	-	-	-	-	-	-	-
P11	V23	GND	ground	GND	I	-	-	-	-	-	-	-
P13	W1	GND	ground	GND	I	-	-	-	-	-	-	-
P14	AA19	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
P18	W7	GND	ground	GND	I	-	-	-	-	-	-	-
P24	W9	GND	ground	GND	I	-	-	-	-	-	-	-
R1	W13	GND	ground	GND	I	-	-	-	-	-	-	-
R2	Y12	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
R3	W17	GND	ground	GND	I	-	-	-	-	-	-	-
R4	AA2	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
R5	AA4	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
R6	W20	GND	ground	GND	I	-	-	-	-	-	-	-
R7	W22	GND	ground	GND	I	-	-	-	-	-	-	-
R9	AA6	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
R10	W23	GND	ground	GND	I	-	-	-	-	-	-	-
R11	Y2	GND	ground	GND	I	-	-	-	-	-	-	-
R12	AB14	VDDIODDR	power	DDR_VREF	I	-	-	-	-	-	-	-
R13	AB19	VDDIODDR	power	VDDIODDR	I	-	-	-	-	-	-	-
R19	Y4	GND	ground	GND	I	-	-	-	-	-	-	-
T1	AA11	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
T2	Y7	GND	ground	GND	I	-	-	-	-	-	-	-
T5	Y10	GND	ground	GND	I	-	-	-	-	-	-	-
T6	AB1	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
T8	AB13	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
T9	Y14	GND	ground	GND	I	-	-	-	-	-	-	-
T10	AC2	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
T11	Y20	GND	ground	GND	I	-	-	-	-	-	-	-
T12	AA1	GND	ground	GND	I	-	-	-	-	-	-	-
T15	AA7	GND	ground	GND	I	-	-	-	-	-	-	-
T21	AA13	GND	ground	GND	I	-	-	-	-	-	-	-
U1	AA14	GND	ground	GND	I	-	-	-	-	-	-	-
U2	AD1	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
U3	AB2	GND	ground	GND	I	-	-	-	-	-	-	-
U4	AE2	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
U5	AF1	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
U7	AF7	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
U8	AB4	GND	ground	GND	I	-	-	-	-	-	-	-
U9	AF12	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
U10	AB12	GND	ground	GND	I	-	-	-	-	-	-	-
U12	AB18	GND	ground	GND	I	-	-	-	-	-	-	-
U14	AC1	GND	ground	GND	I	-	-	-	-	-	-	-
U16	AC31	GND	ground	GND	I	-	-	-	-	-	-	-
U18	AD2	GND	ground	GND	I	-	-	-	-	-	-	-
V1	AG2	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
V2	AE1	GND	ground	GND	I	-	-	-	-	-	-	-
V4	AF2	GND	ground	GND	I	-	-	-	-	-	-	-
V6	AG6	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
V7	AF6	GND	ground	GND	I	-	-	-	-	-	-	-
V9	AF8	GND	ground	GND	I	-	-	-	-	-	-	-
V20	AF9	GND	ground	GND	I	-	-	-	-	-	-	-
V22	AF10	GND	ground	GND	I	-	-	-	-	-	-	-
W1	AF11	GND	ground	GND	I	-	-	-	-	-	-	-
W2	AG8	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
W3	AF13	GND	ground	GND	I	-	-	-	-	-	-	-
W5	AG1	GND	ground	GND	I	-	-	-	-	-	-	-
W6	AG10	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
W7	AG11	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
W9	AG13	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
W11	AG7	GND	ground	GND	I	-	-	-	-	-	-	-
W15	AG9	GND	ground	GND	I	-	-	-	-	-	-	-
W23	AG12	GND	ground	GND	I	-	-	-	-	-	-	-
Y1	AH1	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
Y2	AG19	GND	ground	GND	I	-	-	-	-	-	-	-
Y4	AK5	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
Y5	AG26	GND	ground	GND	I	-	-	-	-	-	-	-
Y6	AK7	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
Y7	AH2	GND	ground	GND	I	-	-	-	-	-	-	-
Y8	AK11	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
Y9	AJ1	GND	ground	GND	I	-	-	-	-	-	-	-
Y10	AK13	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-

Table 9-1. Pin Description⁽¹⁾⁽²⁾ (continued)

427-pin TFBGA	442-pin TFBGA	Power Rail ⁽³⁾	I/O Type ⁽⁴⁾	Primary		Alternate		PIO Peripheral				Reset State ⁽⁵⁾
				Signal	Dir	Signal	Dir	Func	Signal	Dir	I/O Set	
-	AL4	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
-	AL6	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
-	AL8	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
-	AL10	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
-	AL12	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
-	AL14	DDRM_VDD	power	DDRM_VDD	I	-	-	-	-	-	-	-
-	AK2	GND	ground	GND	I	-	-	-	-	-	-	-
-	AK4	GND	ground	GND	I	-	-	-	-	-	-	-
-	AK6	GND	ground	GND	I	-	-	-	-	-	-	-
-	AK8	GND	ground	GND	I	-	-	-	-	-	-	-
-	AK10	GND	ground	GND	I	-	-	-	-	-	-	-
-	AK12	GND	ground	GND	I	-	-	-	-	-	-	-
-	AK14	GND	ground	GND	I	-	-	-	-	-	-	-
-	AL1	GND	ground	GND	I	-	-	-	-	-	-	-
-	AL3	GND	ground	GND	I	-	-	-	-	-	-	-
-	AL5	GND	ground	GND	I	-	-	-	-	-	-	-
-	AL7	GND	ground	GND	I	-	-	-	-	-	-	-
-	AL9	GND	ground	GND	I	-	-	-	-	-	-	-
-	AL11	GND	ground	GND	I	-	-	-	-	-	-	-
-	AL13	GND	ground	GND	I	-	-	-	-	-	-	-
-	AL16	GND	ground	GND	I	-	-	-	-	-	-	-
-	AL23	GND	ground	GND	I	-	-	-	-	-	-	-
-	AL31	GND	ground	GND	I	-	-	-	-	-	-	-

Notes:

1. I/Os for each peripheral are grouped into I/O sets, listed in the column "I/O Set". For all peripherals, use I/Os that belong to the same I/O set. Timings can be unpredictable when I/Os from different I/O sets are mixed.
2. When using an I/O line with the Analog-to-Digital Converter (ADC) or with the Analog Comparator Controller (ACC), the PIO line configuration (pull-up, pull-down) programmed before assigning this line to the ADC or ACC peripheral is not modified by this peripheral.
3. In the TFBGA442 package, the following ground signals are internally connected to the GND balls: GND, GNDANA, GNDIN33, GNDUTMI, GNDBAT, GNDDPHY.
4. Refer to I/O Characteristics in the Electrical Characteristics section of the SAMA7D6 Series data sheet for the definition of I/Os.
5. PU=Pull-Up, PD=Pull-Down, HiZ=High Impedance, ST=Schmitt Trigger
6. This is the PIOBU state after VBAT power-up. If programmed to another value, this value is maintained as long as VBAT is not removed.
7. Open-drain output requires an external resistor.

10. Electrical Characteristics

The Electrical Characteristics sections in the SAMA7D6 Series and DDR SDRAM data sheets (see [Reference Documents](#)) apply to this device. Complementary information is provided in the following sections.

The VDDQ, VDD and VDDL power inputs described in the DDR SDRAM data sheets are connected to the SAMA7D6 Series SiP balls called “DDRM_VDD”. Therefore, the requirements placed on the VDDQ, VDD and VDDL power inputs in the “Absolute Maximum Ratings” and “Recommended DC Operating Conditions” sections of these data sheets apply to the DDRM_VDD power inputs.

10.1. Recommended Thermal Operating Conditions

Table 10-1. Recommended Thermal Operating Conditions

Symbol	Parameter	Min	Max	Unit
T _{J_MPU}	Junction temperature range	-40	105	°C
T _{J_DDR}	Junction temperature range	-40	105	°C

Table 10-2. TFBGA427 Package Thermal Characteristics⁽¹⁾⁽²⁾⁽³⁾

Symbol	Parameter	Typ	Unit
R _{JA}	Junction-to-ambient thermal resistance	21	°C/W

Table 10-3. TFBGA442 Package Thermal Characteristics⁽¹⁾⁽²⁾⁽³⁾

Symbol	Parameter	Typ	Unit
R _{JA}	Junction-to-ambient thermal resistance	28.1	°C/W

Notes:

1. $R_{JA} = (T_{J_MPU} - T_A) / P_{MPU}$, where T_A is the ambient temperature and P_{MPU} is the processor power consumption. The DDR SDRAM junction temperature (T_{J_DDR}) is always lower than the MPU junction temperature.
2. According to the JEDEC JESD51-2 standard, with a 2s2p board and 0 m/s air flow.
3. These values are not directly applicable to the board where the device is mounted. As per JEDEC standards, these parameters do not characterize the package itself but rather the package together with the PCB (4-layer or more) and other environmental factors (still air, etc.). For example, in still-air JEDEC-defined R_{JA} measurements, almost 70% of the power generated by the chip is dissipated from the test board, not from the package surfaces.

10.2. Power Sequences

The DDR SDRAM power rail (DDRM_VDD) must be connected to VDDIODDR on the PCB. Refer to Recommended Power Supply Sequencing in the Electrical Characteristics section of the SAMA7D6 Series data sheet.

10.3. System Power Consumption in Applicative Use Cases

Table 10-5 provides the device power consumption in the following conditions:

- $f_{CPU_CLK} = 800$ MHz
- $f_{MCK1} = 200$ MHz
- $f_{MCK2} = 533$ MHz
- $f_{MCK3} = 266$ MHz
- $f_{MCK4} = 400$ MHz
- $f_{MCK5} = 200$ MHz

- $f_{\text{MCK6}} = 200 \text{ MHz}$
- $f_{\text{MCK7}} = 100 \text{ MHz}$
- $f_{\text{MCK8}} = 100 \text{ MHz}$
- $f_{\text{MCK9}} = 100 \text{ MHz}$
- I & D caches enabled
- Use cases run on Linux®
- Ambient temperature: 25°C
- Current consumption is measured as shown in [Figure 10-1](#). Note that the external component current consumptions are counted.

The measurements are provided for typical SAMA7D65D5M, SAMA7D65D1G, SAMA7D65D2G and SAMA7D65D4G devices.

Table 10-4. Use Case Definition

Use Case	Description
1	Audio MP3 decoding and playback on I ² S; MP3 file on USB mass storage
2	SAMA7D6 running as iPerf server
3	Running Bonnie++ on USB mass storage
4	SAMA7D6 downloads a file from GMAC0 and copies this file to USB mass storage
5	Test pattern display using mode test (720x1280 DSI panel @ 60 Hz)
6	Linux idle

Figure 10-1. Current Measurement for Applicative Use Cases

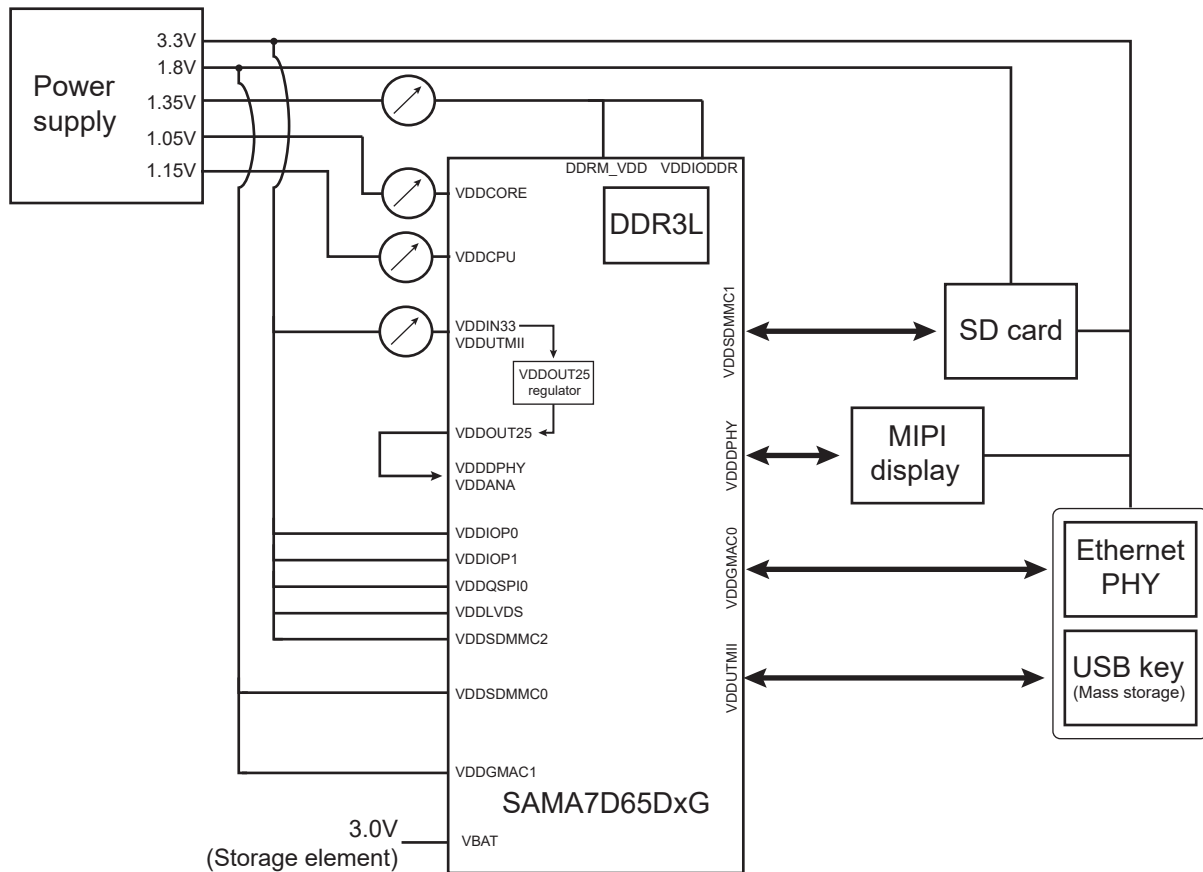


Table 10-5. Power Consumption in Applicative Use Cases

Use Case	Power Consumption (mW)				
	VDDCORE 1.05V	VDDCPU 1.15V	VDDIODDR 1.35V	VDDIN33 VDDUTMII 3.3V	Total
1	181	43	97	181	501
2	204	117	185	224	730
3	195	156	123	176	650
4	183	63	120	173	538
5	206	29	113	197	545
6	189	41	96	164	490

10.4. Power Consumption in Idle and Ultra-Low Power (ULP0, ULP1) Modes with SDRAM in Self-Refresh

For a complete description of how to enter and exit ULP0 or ULP1 mode, refer to the SAMA7D6 Series data sheet (see [Reference Documents](#)).

Table 10-6. Typical Power Consumption in Idle, ULP0 or ULP1 Mode on VDDIODDR and DDRM_VDD

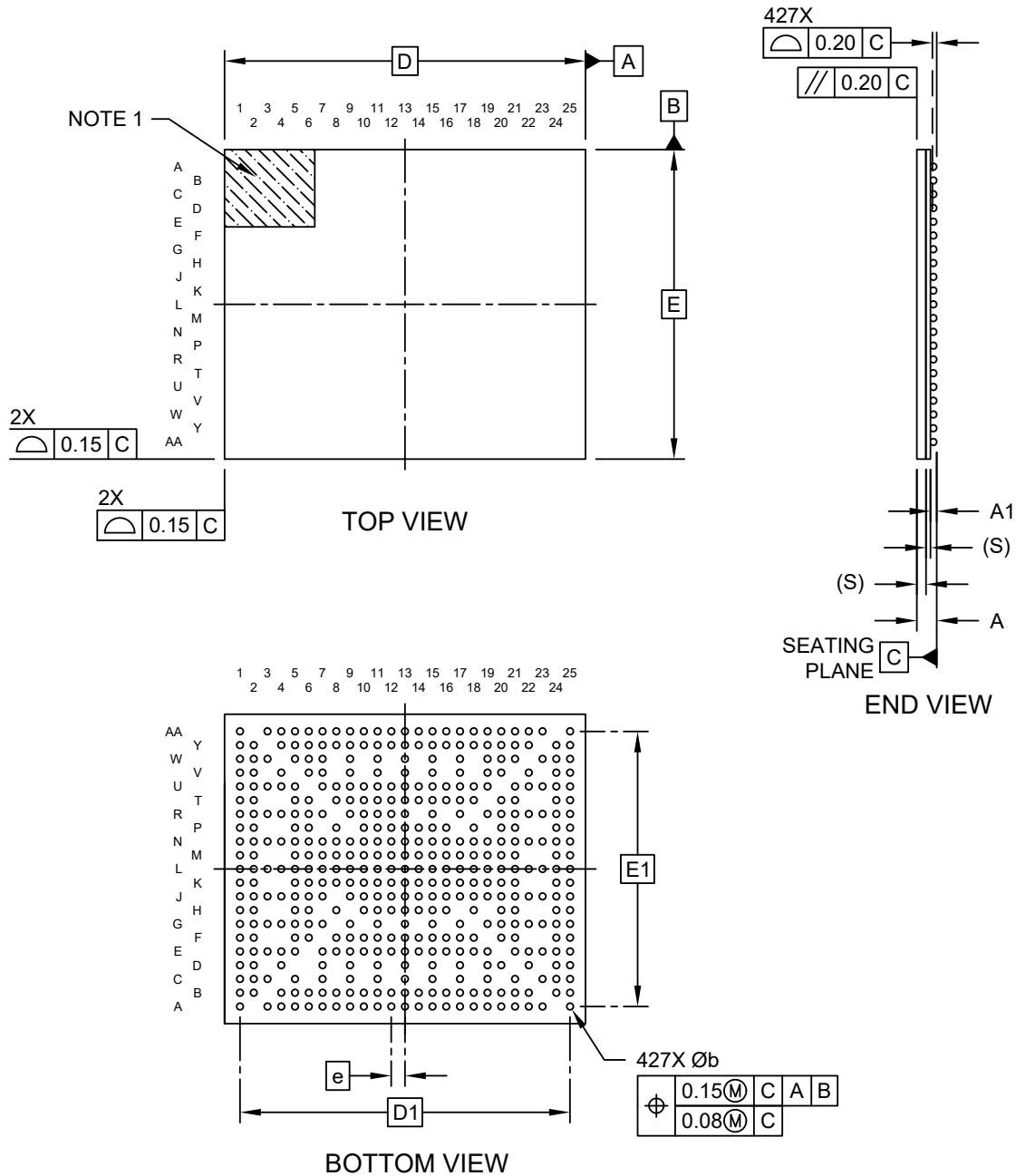
VDDIODDR DDRM_VDD (V)	T _J = -40°C	T _J = 25°C	T _J = 50°C	T _J = 70°C	T _J = 85°C	T _J = 105°C	Unit
1.35	3.7	4.3	5.1	6.3	7.7	10.9	mW

11. Mechanical Characteristics

11.1. 427-Ball TFBGA Mechanical Characteristics

427-Ball Thin Fine-Pitch Ball Grid Array (4UB) - 21x18x1.2 mm Body [TFBGA]

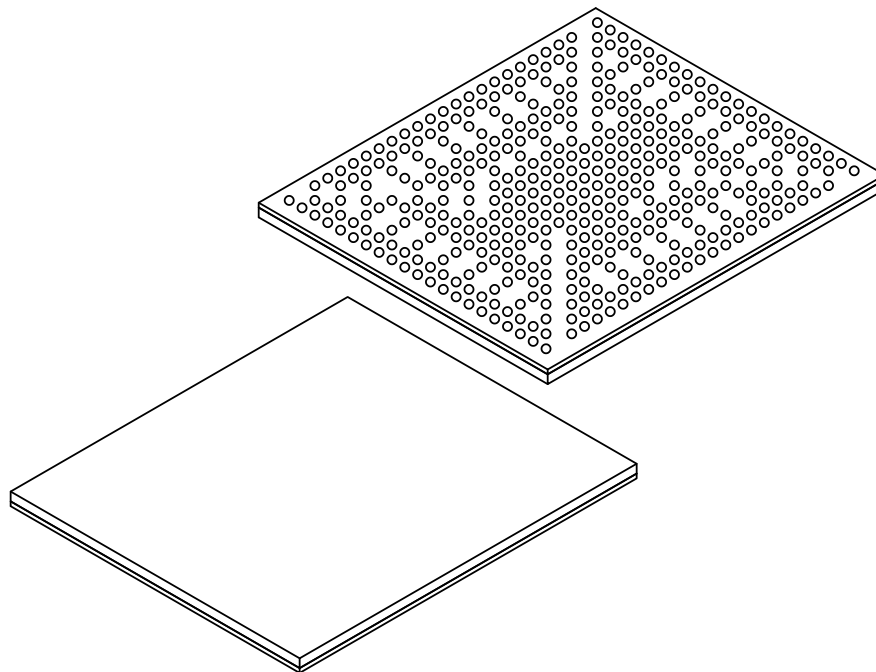
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Microchip Technology Drawing C04-21537 Rev A Sheet 1 of 2

427-Ball Thin Fine-Pitch Ball Grid Array (4UB) - 21x18x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	427		
Pitch	e	0.80 BSC		
Overall Height	A	–	–	1.20
Ball Height	A1	0.27	–	0.37
Mold Thickness	M	0.53 REF		
Substrate Thickness	S	0.26 REF		
Overall Length	D	21.00 BSC		
Ball Array Length	D2	19.20 BSC		
Overall Width	E	18.00 BSC		
Ball Array Width	E2	16.00 BSC		
Ball Width	b	0.38	–	0.45

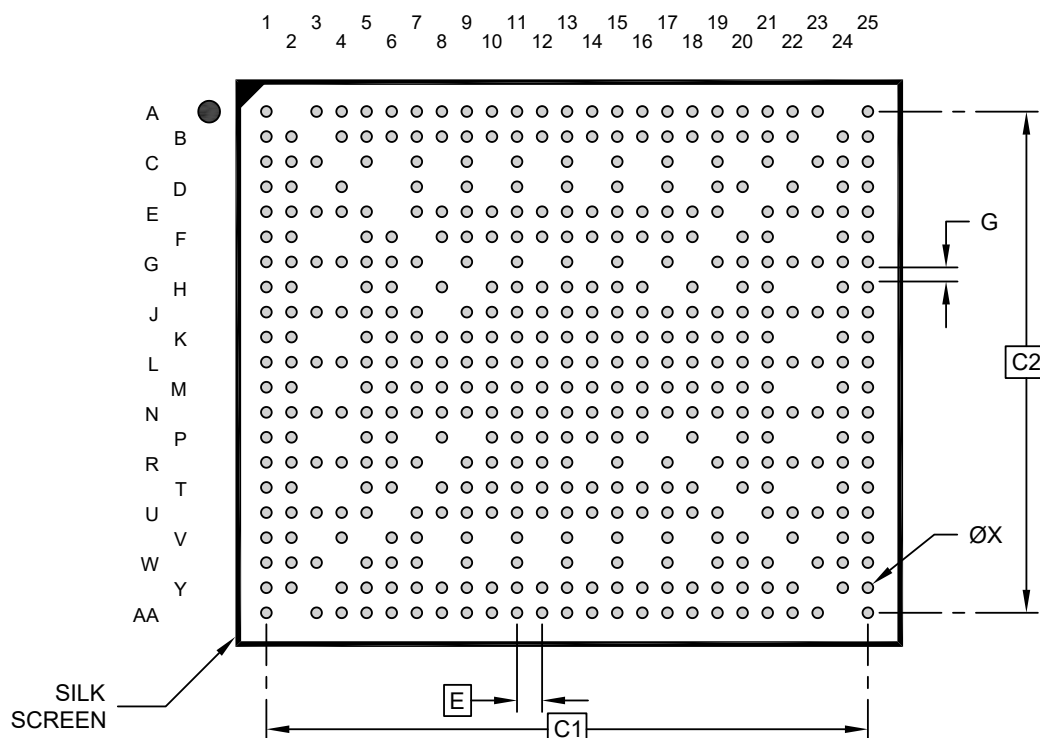
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-21537 Rev A Sheet 2 of 2

427-Ball Thin Fine-Pitch Ball Grid Array (4UB) - 21x18x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

**RECOMMENDED LAND PATTERN**

Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1	19.20 BSC		
Contact Pad Spacing	C2	16.00 BSC		
Contact Pad Width (Xnn)	X			0.35
Contact Pad to Contact Pad (Xnn)	G	0.45		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23537 Rev A

Table 11-1. 427-ball TFBGA Package Characteristics

Moisture sensitivity level	MSL3
----------------------------	------

Table 11-2. Device and 427-ball TFBGA Package Weight

980	mg
-----	----

Table 11-3. Package Reference

JEDEC drawing reference	N/A
J-STD-609 classification	e8

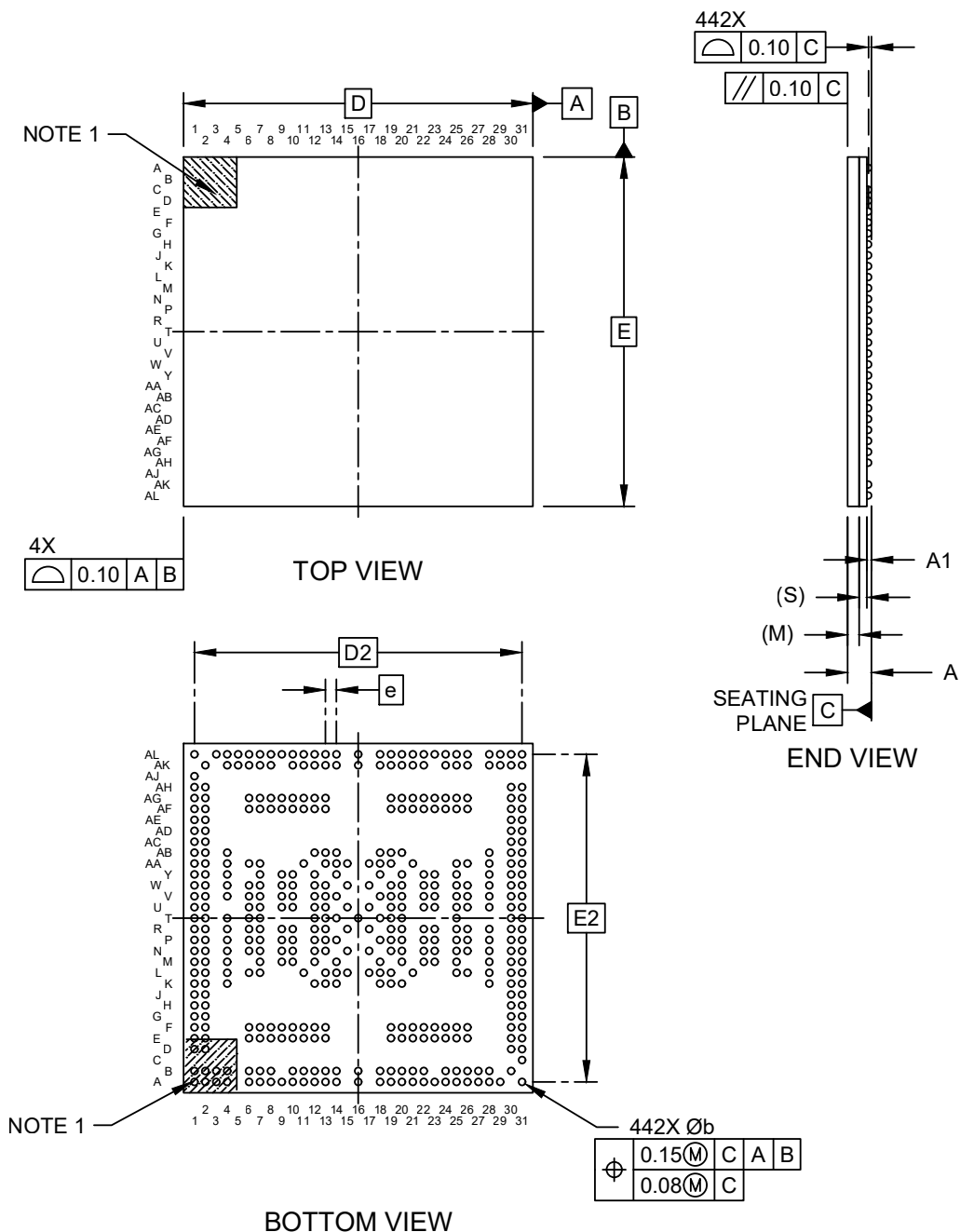
Table 11-4. 427-ball TFBGA Package Information

Ball land	0.450 mm
Nominal ball diameter	0.400 mm
Solder mask opening	0.350 mm
Solder mask definition	Solder Mask Defined (SMD)
Solder	SAC105

11.2. 442-Ball TFBGA Mechanical Characteristics

442-Ball Thin-Profile Fine-Pitch Ball Grid Array (XHB) - 16x16x1.2mm Body [TFBGA]

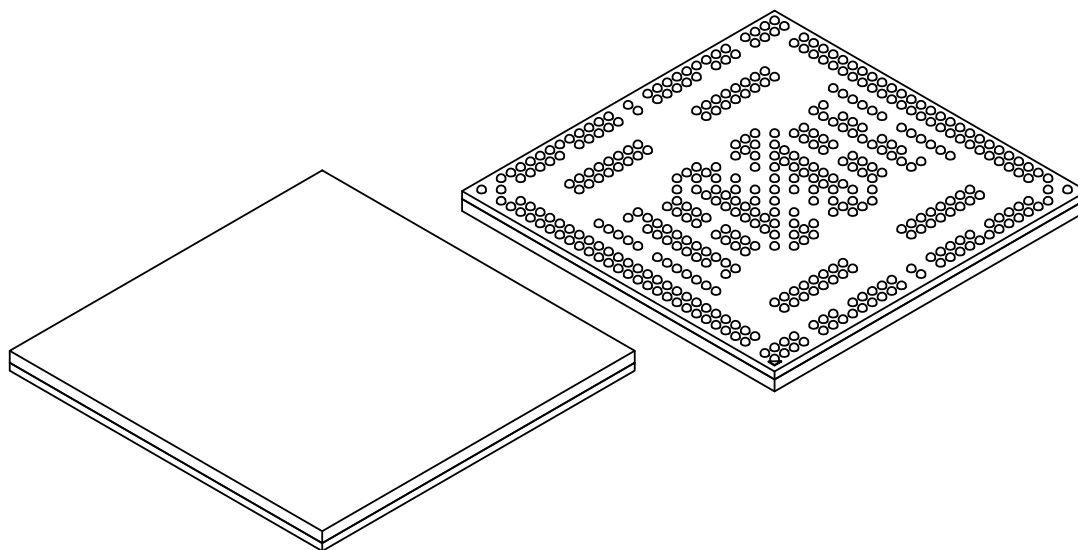
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Microchip Technology Drawing C04-00686 Rev B Sheet 1 of 2

442-Ball Thin-Profile Fine-Pitch Ball Grid Array (XHB) - 16x16x1.2mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	442		
Pitch	e	0.50 BSC		
Overall Height	A	—	—	1.20
Ball Height	A1	0.16	0.21	0.26
Mold Thickness	M	0.53 REF		
Substrate Thickness	S	0.35 REF		
Overall Length	D	16.00 BSC		
Ball Array Length	D2	15.00 BSC		
Overall Width	E	16.00 BSC		
Ball Array Width	E2	15.00 BSC		
Ball Diameter	b	0.27	0.32	0.37

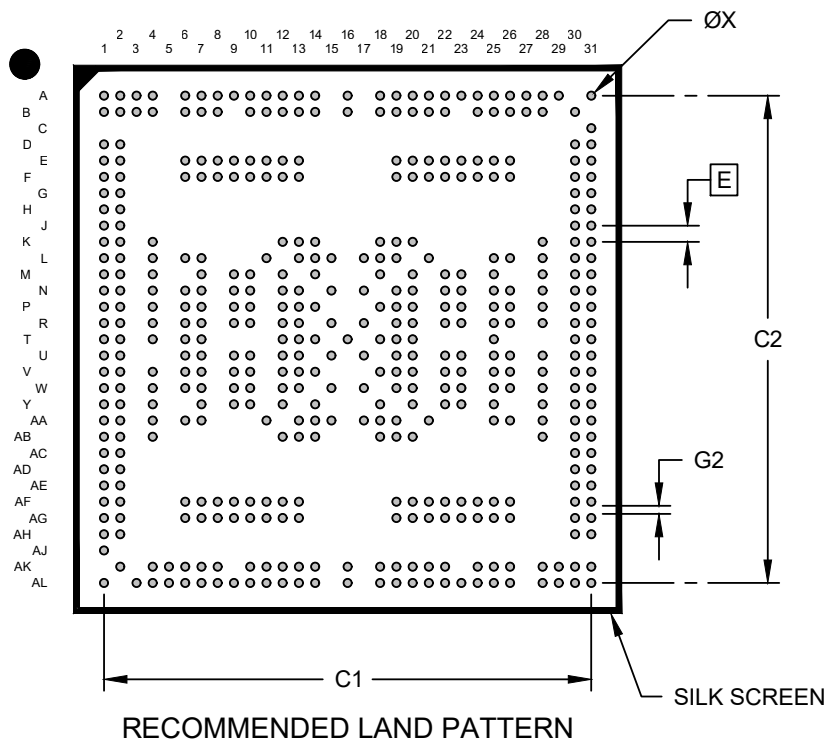
Notes:

1. The A1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M
 BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
 REF: Reference Dimension, usually without tolerance, is for information purposes only.

Microchip Technology Drawing C04-00686 Rev B Sheet 2 of 2

442-Ball Thin-Profile Fine-Pitch Ball Grid Array (XHB) - 16x16x1.2mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Spacing	C1		15.00	
Contact Pad Spacing	C2		15.00	
Contact Pad Diameter (X442)	X			0.25
Contact Pad to Contact Pad	G2	0.25		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. The theoretically exact value is shown without tolerances.
- For best soldering results, please refer to the current industry standard IPC-7093.

Microchip Technology Drawing C04-02686 Rev B

Table 11-5. 442-ball TFBGA Package Characteristics

Moisture sensitivity level	MSL3
----------------------------	------

Table 11-6. Device and 442-ball TFBGA Package Weight

690	mg
-----	----

Table 11-7. Package Reference

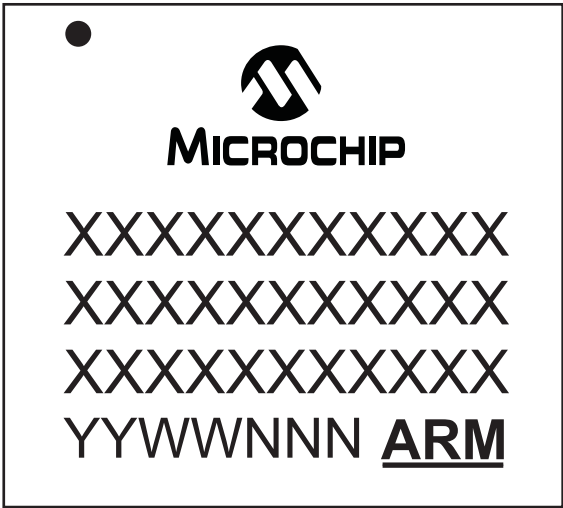
JEDEC drawing reference	N/A
J-STD-609 classification	e8

Table 11-8. 442-ball TFBGA Package Information

Ball land	0.400 mm
Nominal ball diameter	0.320 mm
Solder mask opening	0.275 mm
Solder mask definition	Solder Mask Defined (SMD)
Solder	SAC105

12. Marking

Top marking follows the scheme below:



with possible values:

Line	Description	Values
1	Company logo	Microchip logo
2	Company name	Microchip
3	Device name	SAMA7D65Dxx
4	Temperature code / Packaging code, JEDEC symbol	V/4UB  V/XHB 
5	Not used	-
6	Lot traceability, Arm logo	YYWWNNN ARM

13. Revision History

13.1. Rev. E - 02/2026

Throughout: added content related to the fine pitch devices (SAMA7D65DxG(T)-V/XHB)

13.2. Rev. D - 08/2025

Throughout: added content related to the 4-Gbit device (SAMA7D65D4G)

Updated [Reference Documents](#)

13.3. Rev. C - 07/2025

Throughout: added content related to the 512-Mbit device (SAMA7D65D5M)

13.4. Rev. B - 12/2024

Complete data sheet

13.5. Rev. A - 06/2024

Preliminary issue

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Product Page Links

[SAMA7D65D1G](#), [SAMA7D65D2G](#), [SAMA7D65D4G](#), [SAMA7D65D5M](#)